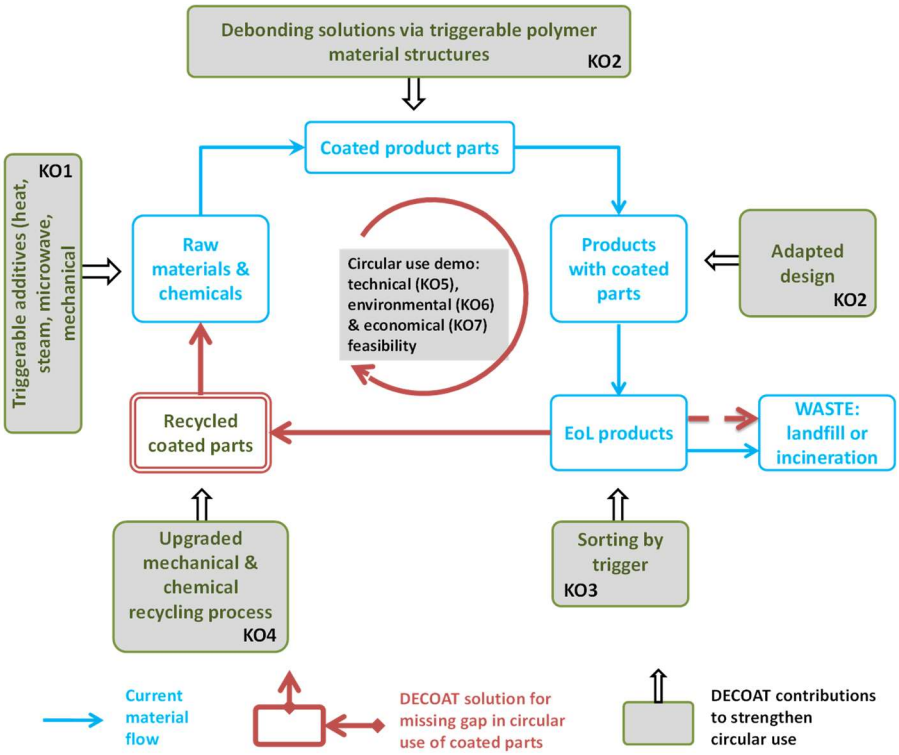
1 Excellence

'You can't solve a problem on the same level it was created. You have to rise above it to the next level'
(Albert Einstein, scientist)

1.1 Objectives

The last decade has experienced a strong increase in technological developments for improved material functionality. One of these involves surface treatments, which employ complex (multilayer) structures designed to provide multifunctionality in a single solution. Within this text, for the sake of simplicity, we will refer to these complex multilayer surface treatments as **'coatings'. This definition includes functional and performance coatings on textiles, paints on plastics, foil lamination and adhesion layers for plastics parts** (polymer-polymer and polymer-metal connections). These 'coatings' have a wide range of advantages and benefits but currently cannot be recycled because of their hybrid nature.



The main goal of DECOAT is to enable the circular use of coated parts by developing novel smart polymer material systems, and their corresponding recycling processes, currently a missing link – see figure. **We will demonstrate the project results via concrete use-cases for outdoor gear, household electronics and automotive parts**. A systemic approach for delivering breakthroughs all along the value chain will be followed, see figure.

To reach this ambitious goal and to maximise the impact of our DECOAT project on the textile and plastics sector, we define the following concrete key objectives (KO):

- **KO1. Development of a toolbox of triggerable materials for debonding on demand** suitable for use on textiles and plastics. Result: 'coatings'[1] that can be removed by various triggers, e.g. thermal via infra-red, heat or steam (targeted temperatures 100-150°C), chemical treatment or 'microwaves'[2].
- **KO2. Development and design of triggerable debonding coating solutions** for 'disassembly' of parts or objects and for 'removal' of the coating from the multilayer structure. Focus is on acrylates, polyurethanes and PVC materials. Result: *generic disassembly and removal solutions* for punctual connections (0D), linear joints (1D) and planar coatings (2D).
- **KO3. Automation of sorting** according to the required trigger. Result: pilot line enabling to sort the coated textiles and plastics based on the separation trigger incorporated.
- **KO4. Integration of debonding systems in a continuous pilot plant recycling line for coated parts.** Result: pilot line enabling the recyclability of coated parts (flexible and rigid). The system will integrate the following elements: sorting and grinding, the novel debonding system, mechanical recycling and melt extrusion, all steps running in a continuous way.
- **KO5. Demonstration via use cases** (outdoor gear, household electronics and automotive parts) using the pilot recycling plant and the adapted coating processes during manufacturing. Result: circularity demo of the use case products. Technology and processing for the use cases brought to TRL 5, accompanied by a plan for upscaling & industrialisation.

1 **'coatings'** refers here to **coatings, paints and adhesives** as defined above and as will be used for simplicity throughout this whole document
2 **'microwaves'** will be the most likely option (although it could also be another form of electromagnetic heating like Rf heating -see also section 1.4.1) and as such will be used for simplicity throughout this whole document

- **KO6**. **Assessment of environmental impact and implementation of safety-by-design** (safe product, safe production and safe use and recycling). Result: full LCA study for the debonding concepts; reduced uncertainties and risks for human and environmental safety of DECOAT solutions.
- **KO7. Exploitation and business plans** for the DECOAT results, including for the use cases manufacturing guidelines to ensure implementation at industrial scale. Further, for the more generic results (design-material concepts) a thorough analysis will be done for wide scope exploitation within the textile and plastics sectors. Result: tailored business plans for DECOAT (company) partners, public research roadmap and a strategy to further exploit and develop the generic DECOAT results, as a basis for implementing the market creating potential of the DECOAT approach.
- **KO8. Dissemination and communication among relevant European communities.** The main target is the European materials and recycling community, supplemented by the wider stakeholders group, both for getting extra feedback during project implementation and for promoting our results. Result: raise awareness, engage for collaboration and adaptation/integration of the project developments in applications beyond those covered within DECOAT.

1.2 Relation to the NMBP work plan

The DECOAT products will be designed to enable separation (and thus recycling) by incorporating smart additives in the adhesive that are activated by a specific trigger (e.g. heat, humidity, microwave, chemical, mechanical). Additionally, an efficient sorting process correlated with the trigger and a pilot recycling plant will be developed to illustrate circular use. In this fashion, **DECOAT will directly respond to two of the three types of materials design objectives put forth in the call scope**: '**Design of polymer material structures with intrinsic sorting/recycling abilities**' (bullet point 1) and '**developments of separation and recycling technologies**' (bullet point 3).

NMBP-26-2018 call - challenge & scope	DECOAT response
*products **with smart intrinsic recycling and/or sorting abilities***	Three product families using multilayer coated parts are targeted: backpacks, household electronics and automotive parts. Each of them will be made from multi-layered materials that allow debonding for recycling and that can be sorted via the trigger mechanism. These are representative for a wide range of coated and painted textiles and plastic based consumer goods.
harmonise with** circular economy **principles	The 3 basic circular economy principles are[3]: (i) Design out waste and pollution; (ii) Keep products and materials in use; (iii) Regenerate natural systems. DECOAT will enable coated, hybrid and painted materials to harmonise with principles (i) and (ii)
*help industry **to better match the EU** environmental targets **at the same time as improving their** competitiveness*	DECOAT targets the re-use and recycling of waste materials from households to increase to a minimum of 50%[4]. We will mainly contribute for 'plastics'. DECOAT will also contribute to the EU environmental targets for the automotive sector[5].
design of** polymer formulations with smart additives**,	DECOAT will make use of a toolbox of smart additives: microwaveable additives, microcapsules or bonding/ debonding adhesives for separation.
***developments of** separation and recycling technologies*	DECOAT will tailor microwave technology, heat, steam and dissolving to enable separation of the coating from the textile or plastic.
demonstrate the** actual circular use **of such materials through re-processing of recycled products	A pilot will be developed for the use cases to demonstrate the circular use of the materials. Both closed-loop (for same application) and open-loop (across applications but of same value level) recycling will be considered.
evaluation of** properties of re-processed products **in an industrial environment	Properties of the use case products made with the reprocessed materials will be evaluated by the RTDs and by the DECOAT manufacturers (DECOAT end-users) according to their normal testing conditions.
full Life Cycle Assessment** (LCA) **of the material production and life-cycle	LCA is included in DECOAT activities and is a Key Objective. Scope: raw material acquisition and processing, manufacturing, distribution and transport, use phase, end of life, and benefits and loads beyond system boundaries (secondary systems).
Activities should start at TRL3, …achieve TRL5	DECOAT approach fulfils this criterion, see section 'positioning of DECOAT'

3 https://www.ellenmacarthurfoundation.org/circular-economy/overview/concept
4 http://ec.europa.eu/environment/waste/framework/targets.htm
5 2000/53/EC directive "… for all end-of life vehicles, the reuse and recovery shall be increased to a minimum of 95 % by an average weight per vehicle and year."

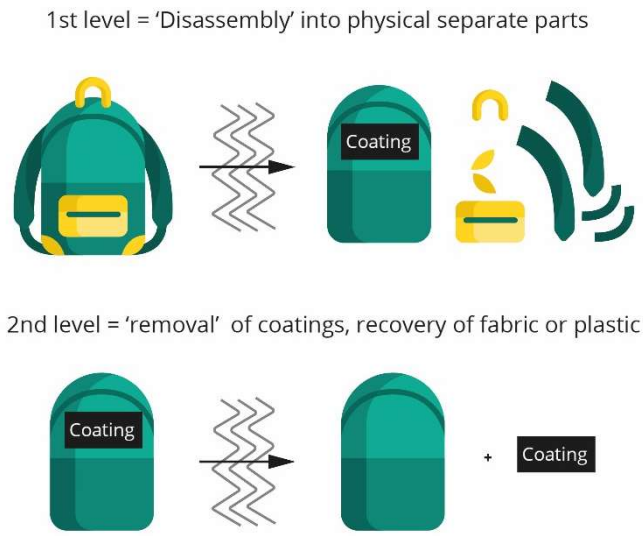
1.3 Concept and approach

(a) Overall concept underpinning DECOAT

The playing field. **Increasing recycling is crucial**, especially for plastics and textiles, as illustrated by the recently published **European strategy on plastics**[6] or the **voluntary engagement of the plastics industry to recycle 50%** by 2040[7]. However large the need and however impressive the targets, **today only 6% of new plastic materials used stems from recycling**[8]. Also within the textile industry the use of recycled material is limited: for clothing only about 1% is closed loop, ca 12% cascade recycling[3]. A great deal of work is underway to improve this (also via European projects), focussing on technical, economic and logistical aspects. As a result, **we assume in the near future well-established networks throughout Europe where waste is collected and brought to local recycling facilities**. At first, this network will deal with the current generation of products and materials that are *not* made for recycling. Consequently, only a small fraction of all materials is being recycled because the cost to the yield and quality ratio is too high compared with the one of virgin materials. Therefore, the next steps are **products designed for recycling and made of intrinsically recyclable materials**. This concept is being developed, e.g. for high added value applications like aerospace. However, for vast application areas this principle is not very advanced yet.

The identified need. **DECOAT will implement a solution for a specific class of hard to recycle items: textiles and plastics parts with (multilayer) 'coatings'**. For plastics, these can be found eg in automotive, construction and electronics consumer goods. For textiles, typical uses are for technical textiles (outdoor gear, protective clothing, automotive applications and construction) and interior textiles (e.g. upholstery, mattress covers or curtains). The coating can be added for performance or aesthetics reasons. Coating, paints or adhesives have to possess excellent adhesion because of the durability of the coated product, and as such cannot easily be removed. Consequently, **full objects are discarded, even though the coating often only comprises some % of the total weight**. With DECOAT we will reduce the amount of waste material to the specific adhesive. This holds for end-of-life products but it is also one of the types of production scraps that are not efficiently recycled yet. Within this larger topic of debonding **an emerging specific challenge is how to deal with scrap and end-of-life parts containing integrated electronics** (from smart textiles, smart plastics and wearables).

DECOAT proposed solution: enable the recycling of coated parts, which refers to a two-level approach: (i) **'disassembly'** is separating the connections among the different physical parts; (ii) '**debonding**' or **'removal'** refers to the removal of coatings or tie layers from the fabric or plastics object. This 2-step approach is illustrated in the figures for a backpack. These two steps can be performed separately (or simultaneously in some cases) during the recycling process. DECOAT triggerable separations mechanisms will be suited for both steps and fill the missing gap for circular use of coated parts (see sketch in section 1.1). Within DECOAT, we will focus mostly on the 2nd level, i.e. the removal of coatings. But, the same principle of triggered debonding enables the disassembly step. So, DECOAT developments are relevant for both steps and we will refer to the processes as 'debonding'.



Interdisciplinary aspects: DECOAT combines expertise and knowledge from several disciplines: **polymer chemistry, physics, coatings and paints, textiles and plastics processing, recycling, design, and environmental science**. This is needed and allows tackling the interdisciplinary issues required for reaching circularity. One such key issue is to **tune the existing knowledge on triggerable materials to industrial requirements for coated textiles and plastics**.

Stakeholder involvement: interaction with the stakeholders is deemed crucial, especially with the waste actors. Therefore, ISWA (NL) has been added as partner, a worldwide association of solid waste actors that promote sustainable waste management and perform dissemination and policy work. For further strong interaction with a wide group of stakeholders, the **DECOAT Advisory and Support Group (ASG)** has been established, containing 17 members from coating and recycling companies, industry associations, a public authority, standardisation bodies and a publisher, see also section 3.2.1.1f and section 4 for a detailed description.

6 http://ec.europa.eu/environment/circular-economy/pdf/plastics-strategy.pdf
7 https://www.plasticsconverters.eu/single-post/2018/01/16/European-Plastics-Industry-works-towards-50-plastics-waste-recycling-by-2040
8 https://ec.europa.eu/commission/sites/beta-political/files/plastics-factsheet-industry_en.pdf

Societal engagement: DECOAT results will **create a new market of recycling coated textiles and plastics**, which will lead to job creation. Of special interest is (partial) disassembly which enables repair, e.g. in small local workshops.

Positioning of DECOAT: Main technology elements, starting and targeted TRL are summarized in the table below:

Technology element – assessment of status	Start	Target
Microwaveable (MW) additives **exist but manufacturing and design methodologies need to be developed for integration in coatings and paints for the DECOAT applications. Rely on triggering of MW additives for fast curing thermoset resins and polyester annealing, which is well-established.**	3	5
Microcapsules for debonding coatings: **concepts based on heat, steam, microwaves are shown at lab scale (CTB and NTUA). R&D needed to validate debonding in relevant environments.**	3	5
Bonding/debonding adhesives**: industrially available (e.g. INDAR) for applications with metals and ceramics but needs to be transferred to polymeric/textile materials**	4	5-6
CreaSolv® **process was developed for recycling of soluble thermoplastics. It will be applied to dissolve polymeric coatings from textiles and polymer surfaces. After separation, the coating polymer can be recovered from the solvent. Suitable solvents need to be developed for specific coating chemistry.**	3	5
Trigger assisted mechanical recycling**: mechanical recycling as established in paper industry will be modified to treat plastic parts and textiles that contain coatings and/or paints by incorporating additional elements (debonding system & sorting technology) and redesign other ones (grinding & drying system)**	4	5-6
Automatic sorting technology**: is used in the plastic recycling sector, based on a number of monitoring technologies. The right technology is to be selected and calibrated for novel DECOAT triggerable materials.**	3	5

Summarized: we start from technologies with TRL 3 (-4) and target TRL5 (-6) by the end of DECOAT. As much as possible **conventional processing technologies will be used**, which eases scaling up. We will continuously monitor the performance of the technologies along the project to ensure reaching TRL 5, as first steps towards future commercialization.

Linked (inter)national research and innovation actions.

Project Acronym (type of project) - Link with DECOAT	Partner (C=Coord)
EcoMeTex (FP7**)** - Separation layer for carpets based on expandable microspheres, will be further developed as one of the separation technologies	CTB
RADIACAM (UK) - Microwaveable yarn for disassembly of mattresses and car seats	CTECH
MARKERS (BE) – Markers for better sorting towards recycling	CTB (C)
RECYCOAT (BE) – recycling of coated materials, eg separation by means of steam or infrared using additives	CTB (C)
FiberEUse (H2020) – Large scale demonstration of new circular economy value-chains based on the reuse of end-of-life fibre reinforced composites	MAIER
SMARTFAN (H2020) - Development of smart composites with self-healing capsules and carbon nanomaterials/ smart recycling activities	NTUA
BYEFOULING (FP7) - Synthesis, functionalization and up-scaling of nanoparticles for marine coating applications. Encapsulation based on Layer Double Hydroxides (LDH), Carbon Nanotubes (CNTs) and Polyoligomeric silsesquioxanes (POSS) for controlled release of antifouling compounds	NTUA
LORCENIS (H2020) - Synthesis of carbonaceous nanomaterials and super absorbent polymers[9]	NTUA
SHELL (GR) - Synthesis of carbon nanotubes and nano-/micro capsules for coating applications	NTUA
INNOVIP (H2020) – Solvent based separation of composite films applied in vacuum isolation panels (VIP) in order to recycle film components during End of life treatment.	IVV
FORCYCLE (DE) – CreaSolv® process has been used to separate polymer/polymer and polymer/metal composites by dissolving one polymeric component of the composite	IVV (C)
FOURMI (FR): Development of thermo-strippable paints for the selective paint removal of aircrafts	RESCOLL

9 Kanellopoulou I.A., Karaxi E.K., Karatza A., Kartsonakis I.A.., Charitidis C.A, Hybrid superabsorbent polymer networks based on crosslinked poly (methacrylic acid) encapsulated with Si02 for novel applications, 5th International Conference of Engineering Against Failure, 20 - 22 June 2018 - Chios Island, Greece

Wear2™ (UK) – The innovative Wear2 technology provides the capability to easily disassemble garments with the application of a simple end-of-life treatment using MW technology.	CTECH
REBIOFOAM (FP7) – The development of a flexible and energy-efficient pressurised microwave heating process to produce 3D-shaped renewable bio-polymer foams for a novel generation of transportation packaging.	CTECH
LIBRE (H2020): Novel precursor fibre materials from blends of modified lignin and biopolymers for carbon fibre. Includes use of dielectric susceptors to enhance the heating efficiency using MW or Rf energy.	CTECH, CTB
ECODISM (FP7) - development of debondable glazing adhesives for ELVs	RESCOLL
ECOBULK (H2020) – Circularity in automotive sector (among others)	IRIS
URBANREC (H2020) New approaches for the valorisation of URBAN bulky waste into high added value RECycled products	AIMPLAS(C) CTB
EXTRUCLEAN (LIFE+): Demonstrative Project for the removal of hazardous substances during the recycling process of polyethylene hazardous waste packages, employing supercritical carbon dioxide (sc-CO2)	AIMPLAS(C)
eCircular - The EIT Climate KIC eCircular Flagship programme will accelerate the circularity of plastic-based material systems and dematerialisation of plastic demand, with the vision of a reaching a carbon-neutral material system by 2050. The Flagship thrives for digital innovations for waste prevention and their up-scaling.	ECOMAT

(b) Overall Methodology:

For describing the general DECOAT methodology, we group the DECOAT activities according to lab-scale, pilot scale and demonstration. The supporting activities are mentioned separately.

Materials and material structures R&D at lab scale

'Triggerable materials': R&D on the microcapsules, nano-additives, their integration in coatings, characterisation of their effects when triggered. We will develop a portfolio of techniques, each will be characterised, listing the strengths and weaknesses for the respective application domains. Integration in a coating system, directly in the coating or via a primer.

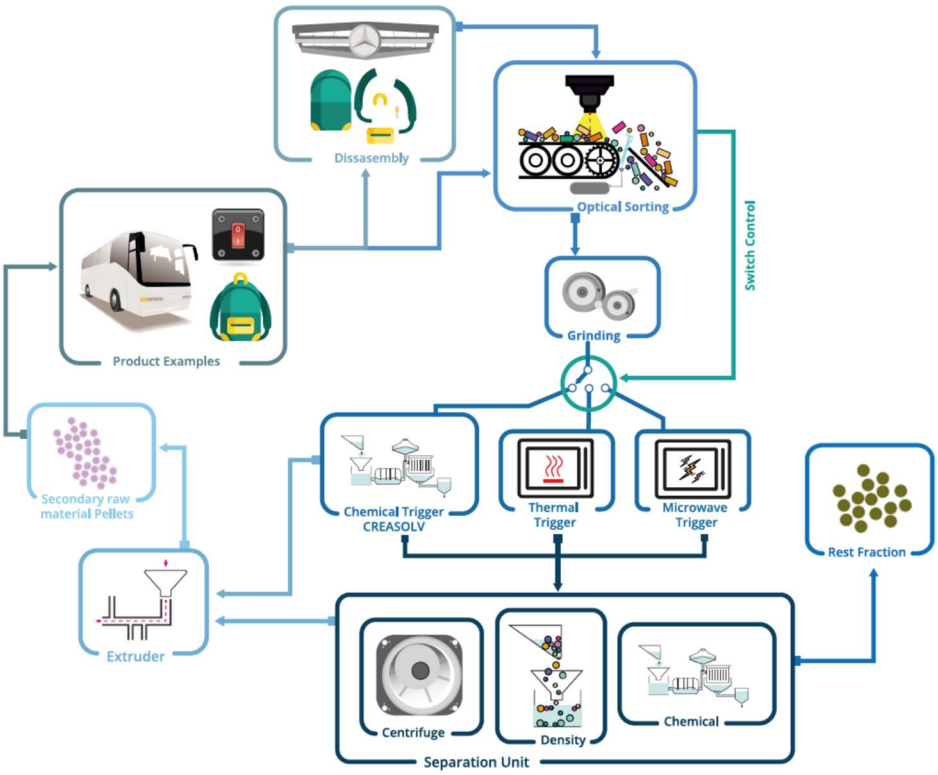
Triggerable debonding coating solutions both for the 'disassembly' of parts ('1st level) and for the 'removal' of the coating from the substrate (2nd level).

Process development at pilot scale:

Sorting technology: Research on sorting to discriminate post-consumer and post production waste based on type of debonding trigger to ensure the materials can be identified & guided to the fitting recycling technique.

Piloting of a recycling line for coated parts: conception, design and building of an actual pilot line, performing test runs on generic coated textiles and plastic parts, validation via the recycling of the use cases prototypes.

Demonstration of circularity via use cases (see sketch): The novel coatings will be applied in the existing processes by the manufacturers/ end-users via three use cases. The (small scale) production of products will undergo the recycling process, the recovered materials will be characterised and re-used. Evaluation of the reprocessed products.



Transversal activities:

Environmental & safety assessment: detailed assessment of environmental (full LCA) aspects and development of an ecodesign tool. Ensure Safety-by-Design implementation via use of non-hazardous substances, e.g. for microcapsules or solvents (EC Regulation No. 1272/2008). Apply proper recycling and waste treatment (Directive 2008/98/EC).
Innovation management: Economic assessment (full cost of ownership, business plans); dissemination: raise awareness from early on to maximise wide uptake of DECOAT results; communication to promote public awareness of circular economy values; focus on interaction with wider stakeholders and with the ASG.
Coordination: efficient project implementation, project risk management.

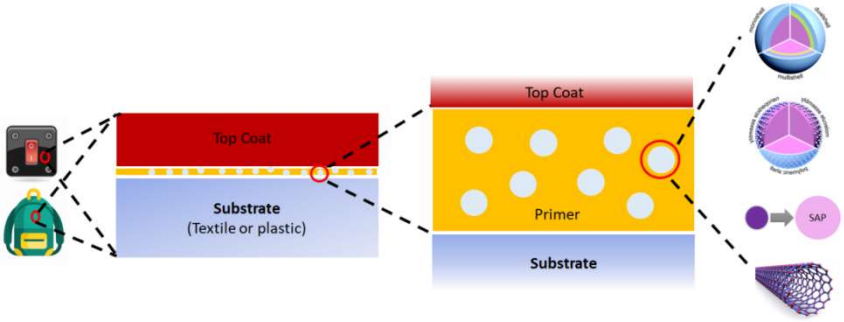
Gender analysis. We will follow **gender driven innovation** and develop **gender neutral demonstrators**. DECOAT partners have a strong focus on women's employment (e.g. at VAUDE 65% women, including the CEO and 33% of executive positions)[10]. Further, our innovation activities will consider the gender dimension and consider the needs, motivations and differences of females where applicable, inspired by how taking gender into account has yielded new research results and sparked creativity[11]. DECOAT will consider the relevance of the gender dimension in communication & training.

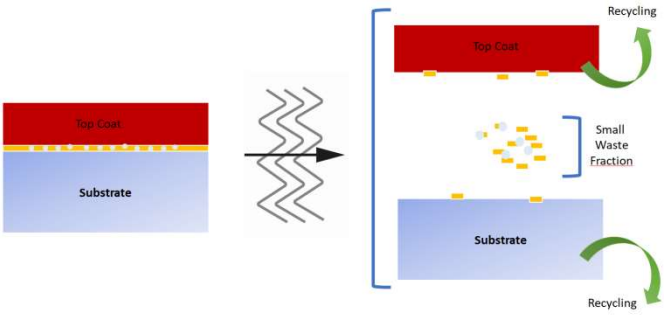
1.4 Ambition

DECOAT aims at overcoming obstacles from current recycling technologies by designing coating formulations with smart additives, which allow adequate sorting, separation and recycling. Developments of separation and recycling technologies will be investigated and new separation technologies utilising novel mechanical and chemical recycling procedures will be validated, to meet the EU's circular economy and environmental targets.

1.4.1 Progress beyond state-of-the-art

Building further on the available knowledge, expertise and patented IP within the DECOAT consortium, we will research and develop a debonding system based on triggerable additives, mechanical recycling, chemical recycling (Creasolv process) and latest coating (monitoring) technology. We will investigate two debonding systems: (i) a single layer with the trigger inside and (ii) a two-layer system consisting of a standard coating and a 'primer' layer (see sketches). Note that 'primer' is used here as common name for tie layer or primer or base layer. For selected combinations of the coating-substrate materials, we will research several triggers and several options for the actual removal of the coating (process conditions, 1-step versus 2-step). The output will be a portfolio of debonding strategies, mentioning for each of the solutions the pros and cons. This novel information will form the basis for our DECOAT demonstration of circular use of coated parts but also be made available to the wider stakeholder community for further use.





In the next paragraphs we describe the concrete **progress beyond the state-of-the-art that DECOAT ambitions for (i) triggerable materials, (ii) optical monitoring for process optimisation and sorting of the coated waste, (iii) actual recycling and (iv) eco-design aspects**.

(i) Triggerable materials and related debonding systems – mainly related to WP1 & WP2 work

Coatings that debond on demand lose their adhesive function when an external trigger is applied. Partner **CTB researched a novel separation layer for textiles by embedding thermally expandable syntactic microspheres** (in FP7 project ECOMETEX). Separation concept was shown on lab scale in boiling water, which is *not* an efficient process as it needs extensive energy (heating, drying). Partner **RESCOLL developed an innovative, patented (see section 1.4.2) interfacial debonding adhesive technology (INDAR™ Inside**) permitting the dismantling of structural aerospace assemblies without damage and was used for selective thermal stripping of aircraft paints. Partner **NTUA** has developed a variety of **'smart' nano- and microcapsules** based on various materials (ceramics, metals, polymers), that react on external stimuli (water, mechanical stresses,

10 Source: http://www.vaude.com/angebote/nachhaltigkeitsbericht/2013_Nachhaltigkeitsbericht_screen_ENGL.pdf
11 Londa Schiebinger (2008). Gendered Innovations in Science and Engineering. ISBN: 978080475814

microorganisms, etc), in projects LORCENCIS and BYEFOULING (see also project table above). Partner **CTECH developed additives for interaction with microwaves**. Based on that, a unique patented (polyester) yarn was developed, which disintegrates during microwaving[12]. This yarn enables easy and inexpensive removal of e.g. zippers (disassembly step).
DECOAT novelty: Using the available IP and expertise, we will develop **triggerable additives** for water-based coating formulations **with a lowered debonding temperature** as compared to existing procedures, although exceeding the processing and working temperature required/set by industrial demo cases. A sufficient difference between the manufacturing, maximal service and the debonding temperature will be ensured and will be realised by improving the heat direction towards the additives, e.g. by heat focussing layers or thermal conductive paints. Thermo-responsive core-shell nanostructured co-polymers, Super Absorbent Polymers (SAP) and novel hybrid microwave susceptible materials (based on metal nanoparticles, silicon carbides and multiwall carbon nanotubes MWCNTs) will be investigated. Separation will be done by heat sources (like steam, microwaves and IR) that are available at industrial level to avoid the need for additional, or only very limited, drying. To reach a clean surface after debonding, we focus on controlled interfacial decohesion. For detection purposes, fluorescent molecules will be added on the "smart" additives (dyes).

The table below gives an overview of the materials to be developed and the polymer coating matrices to which they can be added. Already a first matching between the trigger mechanism and the polymer matrix is made, based on the chemistry of the particles and what is available in literature. As mentioned this is a *first* matching, further investigation within the DECOAT project is needed. Below the table, each of the triggerable materials is discussed, giving more info on their synthesis and the challenges for their use in an actual debonding system.

Triggerable materials	Coating Polymer matrix	Triggers
Thermo-responsive core-shell nanostructured co-polymers Core: e.g. poly(propyl methacrylate) or poly(butyl methacrylate) Shell: e.g. poly(4-vinylpyridine-co-methacrylic acid), ethylene-glycol-bismethacrylate, Functionalised with: organic dye (I), iron oxides (supermagnetic nanoparticles)	Acrylic, PU[13] PVC[14]	Heat / Steam (Temperature)
Super Absorbent Polymers Core: Cross-linked poly(methacrylate-co-ethylene glycol dimethacrylate) Functionalised with: organic dye (II), iron oxides (supermagnetic nanoparticles)	Acrylic Epoxies[15]	Selective Solvents (H_20 or organic solvents)
Microwave assisted materials MWCNTs (decorated with metal nanoparticles or grown on SiO_2 nanoparticles) Susceptors: Metaloxide nanoparticles (e.g. ferroelectric oxide particles), BaTiO3, siliconcarbide, ZnO, silica or alumina Blended with high dielectric polymer	PU[16] PVC[17] Epoxies[18]	MW irradiation
Blowing agents (INDAR) Blowing agents: azodicarbonamide and polycarboxylic acid Blended into formulations for primers, combination with other triggers	Epoxies, PU[19]	Heat / Steam (Temperature)

Thermo-responsive core-shell nanostructured co-polymers

Partner NTUA has vast experience in synthesis and development of tailored microcapsules and nanoadditives with functional properties (currently at TRL3)[20,21,22]. Partner DEVAN has built over the years a strong IP portfolio of microcapsules for various applications[23].

DECOAT novelty: we will make progress on the following topics to realise this triggerable material:

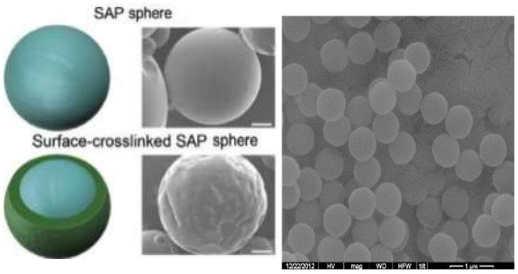
12 Patent WO2013/189956A1, see also www.wear2.com
13 M.Hirose, et al.,The structure and properties of core-shell type acrylic-polyurethane hybrid aqueous emulsions, Progress in Organic Coatings, 31, (1–2) 1997, 157-169
14 A. Gharieh, et al., Preparation of core–shell impact modifier particles for PVC with nanometric shell thickness through seeded emulsion polymerization, Iranian Polymer Journal 23(1), 2013
15 I.A. Kartsonakis, et al., Multifunctional epoxy coatings combining a mixture of traps and inhibitor loaded nanocontainers for corrosion protection of AA2024-T3, Corrosion Science, 85, 2014, 147-159.
16 E. Orgiles-Calpena, et al., Effect of Amount of Carbon Nanotubes in Polyurethane Dispersions, Macromolecular Symposia, 2012.
17 K. Trommer, et al., Processing and Properties of Carbon Nanotube PVC Composites, L. Appl. Chem., 2014, 10
18 J. Zhang, et al., Functional interphases with multi-walled carbon nanotubes in glass fibre/epoxy composites, Carbon, 48, 8, 2010, 2273-2281
19 EP2265681 (B1), EP2519596 (B1), EP2519565 (A1), EP1611217 (B1), EP1814935 (B1)
20 A.C. Balaskas, T. Hashimoto, M. Curioni, G.E. Thomson, Nanoscale 2017 9:5499-5508.
21 A.C. Balaskas, I.A. Kartsonakis, L.-A. Tziveleka, G.C. Kordas, Progress in Organic Coatings 2012 74:418-426.
22 I.A. Kartsonakis, A.C. Balaskas, E.P. Koumoulos, C.A. Charitidis, G.C. Kordas, Corrosion Science 2012 57: 30-41.
23 Patents EP1658131 (B1), US2008193761 (A1), US2016009596 (A1)

Microcapsule core: **selection of polymers with thermo-responsive properties** or inclusion of active ingredients that help delamination of textile layers (eg. poly(propyl methacrylate) or poly(butyl methacrylate).

Microcapsule shell: **selection of suitable** shell components to enclose the core material. **In thermo-responsive core-shell nanostructured co-polymers, a protective thermoplastic shell with tunable glass transition temperature (T_g) and melting temperature (T_m), will be tailored to have a higher T_g than the core's polymer**. The T_g is highly related to segmental polymer chain motion. As polymers warm up, the shell with the high T_g will act as a protective coating for the polymeric core, which will be in melting conditions at elevated temperature. When the ambient temperature surpasses the T_g of the shell, the inner core (material) will have a high chain mobility enabling detaching of the surfaces. At this defined moment the core material is released from the capsule, triggering debonding. On the one hand, the shells should not react with the core material and on the other hand they should have a good chemical compatibility with the surrounding matrices. Examples of possible materials are: melamine resins, polyurea/polyurethanes, polyacrylates. In addition, the shells should provide a good storage for the active ingredients during the entire lifetime of the products, and only rupture/or release the active ingredient when triggered (end of the product life).

Functionalization of the protective thermoplastic shell to increase its compatibility with a wide range of 'coating' formulations, also for INDAR technology (see below). The main materials: PU, PVC, acrylic, epoxies. Of potential further interest: amino, cyanates, isocyanides, hydroxyls, sulphonamides, and phthalates. DECOAT consortium is aware that some forms of these substances have restrictions, so **REACH compliance will be checked** (see also Task 6.1).

The use of the microcapsules in the debonding system: The microcapsule concentration (ca. 5% or less) and cross-linking density of the adhesive will be tailored and optimized, for strong adhesion during product use but also for ease of debonding at end-of-life. The latter will be achieved by exposure to heat as this will melt the shell, and thus activate the core. Homogeneous heat dissipation and activation of the microcapsules will be obtained by adding superparamagnetic nanoparticles (Fe_3O_4). Application of irradiation will locally (=near the microcapsules) raise the temperature[24,25]. **The aim is to develop materials that will be able to create 'soft flexible points' inside the crosslinked adhesive network in the presence of the external stimuli, thus improving detachment.**



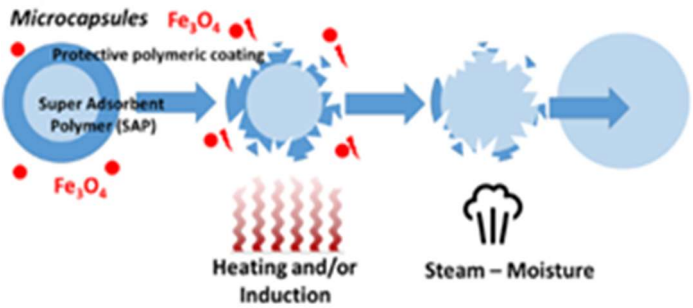
Summarizing, the trigger mechanism for the thermo-responsive core-shell nanostructured co-polymers relies on the different physical properties (T_g) between the core and the shell materials that construct the primer layer. The induction of heat onto the core-shell materials above the T_g of the organic shell, results in the different polymer chain mobility enhancement between the core and the shell. The outcome is the reduction of the attachment between the primer layer and both the top coating and substrate, enabling the detachment of the top coating from the corresponding substrate.

Super Absorbent Polymers (SAPs)

SAPs can absorb and retain extremely large amounts of a liquid relative to their own mass. Water-absorbing polymers, which are classified as hydrogels when cross-linked[26], absorb aqueous solutions through hydrogen bonding with water molecules. In deionized and distilled water, a SAP may absorb 300 times its weight (from 30 to 60 times its own volume) and can become up to 99.9% liquid.

DECOAT Novelty: The SAP materials will exhibit a spherical shape with a **two-layered structure**: the polymeric core will be a lightly crosslinked hydrogel and the outer surface will be a highly crosslinked layer. The water absorption capacity of the SAPs is influenced by the nature and relative amounts of monomers used in the synthesis[27]. In this sense, the regulation of the crosslinker amount (both in the core and shell material) will be the key-property to achieve the **desired degree of water absorbency** for the debonding effect as well as the required protection of the hydrophilic core for obtaining a **stable waterborne formulation**[28]. Clearly, the amount of crosslinker is a decisive parameter for the swelling behaviour of hydrogels and the aim is to achieve a compromise between the crosslinking level and a stable gel structure[29,30].

24Journal of Colloid and Interface Science, Volume 435, 2014, 99-104.
25Journal of Magnetism and Magnetic Materials, Volume 302, Issue 1, 2006, Pages 267-271.
26 Kabiri, K. (2003). "Synthesis of fast-swelling superabsorbent hydrogels: effect of crosslinker type and concentration on porosity and absorption rate". European Polymer Journal. 39 (7): 1341–1348
27Kabiri, K., et al., Superabsorbent hydrogel composites and nanocomposites: A review. Polymer Composites, 2011. 32(2): p. 277-289.
28Chang, S., et al., Multi-scale characterization of surface-crosslinked superabsorbent polymer hydrogel spheres. Polymer, 2018. 145: p. 174-183.
29Goncalves, A.A.L., et al., Synthesis and characterization of high performance superabsorbent hydrogelsusingbis[2-(methacryloyloxy)ethyl] phosphate as crosslinker. Express Polymer Letters, 2016. 10(3): p. 248-258.
30Cipriano, B.H., et al., Superabsorbent Hydrogels That Are Robust and Highly Stretchable. Macromolecules, 2014. 47(13): p. 4445-4452.



The **effect of the pH** on the water absorbency of the SAP spheres will be also investigated. The dispersion of the core-shell SAP materials will be investigated in an aqueous solvent where swelling equilibrium experiments will be performed to study water absorption at different pH. When the pH favours the repulsive forces between the polymeric chains, the water absorbency is increased. On the other hand, when a screening effect of the polar groups of the polymer chains takes place, the repulsive forces between the charged polymer chains are minimized and the water absorbency in greatly decreased. Based on this phenomenon[27], the pH value of the waterborne formulations developed will be selected based on the value where the minimum water absorbency is observed. In this way, the **SAP materials will be in a not swollen state during their incorporation in the coatings and they will be activated only upon the trigger.** The trigger mechanism for the core-shell hydrophilic SAPs relies on the different ability of the core and the shell materials to be swollen under variation of pH. Therefore, under the influence of water with specific pH value, the SAPs can enhance their shape due to the water absorption resulting in the reduction of the attachment between the primer layer and both the top coating and substrate, enabling the detachment of the top coating from the corresponding substrate.

To apply the MW trigger, these structures will be also functionalized with thermally-responsive magnetic nanoparticles (eg Mn-Zn ferrites or Ni or iron oxide) which will be selected based on the Curie temperature that matches the crosslinked layer's melting point[31]. This functionalization will make the shell more susceptible to heating so that the shell can be selectively molten, as described also for the case of the thermo-responsive core-shell co-polymers, allowing the lightly crosslinked hydrophilic core to absorb a large amount of water, causing the swelling and activating the debonding.

To apply the Creasolv process, lipophilic super-absorbent polymers[32] for the absorption of the desirable organic solvents will be prepared by selecting the appropriate compounds both for the polymer chains that control the compatibility with the media and the ionic groups that are dissociable in polymer gels (e.g. alkylammonium salts). Candidate materials include acrylate monomers copolymerized with methacrylates as crosslinkers to yield lipophilic polyelectrolyte gels. The nature of the organic solvent and specifically its polarity (dielectric constant) will influence the swelling behaviour of the SAP materials produced. For this reason, the ionic group content will be adjusted to avoid the formation of physical interactions which will negatively influence the swelling behaviour of the lipophilic SAP. The trigger mechanism for the core-shell hydrophylic SAPs relies on the different ability of the core and the shell materials to be swollen under the presence of an organic solvent. Thus, under the influence of an organic solvent that has enhanced chemical affinity with the shell material, the SAPs can enhance their shape due to the organic solvent absorbtion water resulting in the reduction of the attachment between the primer layer and both the top coating and substrate, enabling the detachment of the top coating from the corresponding substrate.

Hybrid materials reacting to MW triggers

Two forms of dielectric heating are commonly used industrially differing in wavelength of the applied electromagnetic radiation. These are Rf heating using wavelengths at 27 and 40MHz and MW heating using 2.45Ghz and 896MHz. Depending upon the material to be processed, there are advantages and disadvantages of using the different forms of radiation, but the main advantage is the direct heating of the material by the application of the radiation with no external heat transfer or hot surface required. Using Mw and Rf can be done either as a batch system or a semi batch system. In the latter case the materials are placed into a screened cavity and then heated by the application of the desired power for a defined time. This gives some advantages in the design because the system can be enclosed ensuring no leakage of radiation. Materials can be introduced in containers, treated and then removed to be processed further. Continuous processes are also possible for both Rf and MW systems for example conveyers can be used to introduce and remove the materials into the applicator zone.

The main microwave absorption mechanism of a polymer is reorientation of the dipoles when submitted to an electric field, the presence of these groups in the molecule is necessary. Some polymers contain groups with large dipoles, such as epoxies, amino, cyanate, isocyanides, hydroxyls, sulphonamides and phthalates. The materials with the best dipole mobility exhibit high absorption efficiency and the efficiency of the coupling depends on the strength of the dipole, its mobility and mass, as well as its dipole matrix state. The choice of the polymers and the formulation will therefore need to be investigated for an optimal heating effect. By applying electromagnetic radiation to a material, movement of the molecules via dipole interactions with the oscillating electric field is induced, resulting in molecular friction. This leads to dielectric heating of materials.

[31]A. S. Ahmed y R. V. Ramanujan, Curie temperature controlled self-healing magnet–polymer composites, J. Mater. Res., 2015. 30 (07): p. 946-958.
[32]Ono, T., et al., Lipophilic polyelectrolyte gels as super-absorbent polymers for nonpolar organic solvents. Nat Mater, 2007. 6(6): p. 429-33.

Comparison of Rf and MW heating key properties		
Parameter	**Rf**	**MW**
Applied heating efficiency	order of 60%	order of 65%
Uniformity of heating	Good	Can be less uniform
Impedance matching	Can be difficult for certain applications	Easily applied
Energy delivery	Accurate and fast	Less controlled
Remote power sources	Ok	Ok with waveguide transfer to the applicator
Screening to comply with EMC requirements	Easy	More difficult
Tailor heating to materials	Easy	(More) difficult
Common characteristics		
Volumetric heating –no external heat transfer	Dielectric heating does not require physical contact with the product	Low shear transport (conveyorised possible)

Typical susceptors that are used nowadays for MW heating are metaloxide nanoparticles (e.g. ferroelectric oxide particles e.g. $BaTiO_3$, siliconcarbide, ZnO, silica or alumina). **To date, use of microwave heating in carbon-based materials research of is limited**, mainly focused on liquid solutions. However, more rapid and efficient heating is possible in electron-rich solid materials, because the target materials absorb energy of microwaves effectively and exclusively. Carbon-based solid materials are suitable for microwave-heating. Specifically, carbon nanotubes (CNTs) have excellent electrical properties resulting from sp^2 networks of carbon atoms. The number of free electrons which can migrate freely over the whole of the layer is about one per each carbon atom. The free electrons enable CNTs to absorb microwave energy considerably and to be heated effectively by microwaves[33].The tuning feasibility of the permittivity value by changing MWCNTs concentration (1-5%) has been also proven[34]. Other applications like annealing, regeneration and detection based on microwave heating of CNTs have been reported.

<u>DECOAT novelty:</u>

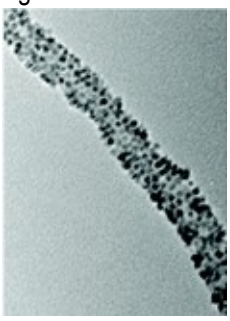
The main idea is the possibility to include susceptors in a primer layer to provide a heating effect to separate the substrate and the top coating. The susceptor may also be part of the actual coating in case of a single layer. The bulk polymer would need to be a low dielectric material to avoid MW heating of the bulk. The aim is to provide selective heating in the area of the susceptor, to either melt (and remove) the primer and thus also the coating on top, or to melt the entire coating itself and remove the coating layer.

For the heating, we will start with MW heating as most likely option. Also Rf will be tested and the possibility to add (as extra) some radiant heating component. In DECOAT, it will be investigated which would be the best energy source, or combination, to use in order to achieve the effect.

To tune the heating and to make it more uniform, we will optimise the % and type of additive. We will also investigate the addition of MW transparent material such as silica or alumina to disperse the particles and moderate the heating effect. Indeed, heating needs to be limited to prevent thermal runaway causing melting or charring of the bulk polymer. Issues might occur if the component contains e.g. metal parts (e.g. fixings/brackets) which could cause arcing or materials fillers that heat strongly in a MW/Rf field. This potential variability of feed would need to be considered, even though we will try to limit the variability via the previous disassembly step and the monitoring and sorting of the waste.

[33]T. Kim, et al., Microwave heating of carbon-based solid materials, Carbon letters Vol. 15, No. 1, 15-24, 2014.
[34]M. Giorcelli, et al., Microwave absorption properties in epoxy resin Multi Walled Carbon Nanotubes composites, 2013 International Conference on Electromagnetics in Advanced Applications (ICEAA), Italy, 9-13 Sept. 2013

We will start making formulations with common susceptors such as small metal particles (nano particles), ferroelectric oxide particles e.g. $BaTiO_3$, silicon carbide or blends with high dielectric polymer. The primer layer or the coating will need a lower melting point compared to the bulk material.



A novelty for targeted MW heating will be the exploration of carbon-based particles, like carbon powder, MWCNT and CNT/MO (Metal Oxide) nanocomposites (see picture on the side[35]). The latter are a new generation with interesting properties that not only combine the properties of CNTs and MO but also hold new properties caused by the interaction between them. Additionally, use of these materials in a polymer coating formulation can overcome some common disadvantages. For example, metal oxide nanoparticles tend to agglomerate because of their dangling bonds. In CNT/MO composites, depositing nanoparticles on the surface of the nanotube prevents their agglomeration. The aggregation of nanotubes in aqueous media is solved by functionalization. As a result, these nanocomposites are attractive for a wide range of applications compared to the isolated nanoparticles because CNTs act as a carrier to stabilize the nanoparticles, maintaining their integrity. This will help the induction of the heat through MW and Rf inside the coating, **since CNTs will work as a thermal conductive network inside the polymer matrix, which should be very beneficial for the debonding functionality**. The result will be the reduction of the attachment between the primer layer and both the top coating and substrate, enabling the detachment of the top coating from the corresponding substrate.

(ii) Automated Sorting technology based on DECOAT triggerable additives – mainly related to WP3 & WP4 work

For the actual recycling of DECOAT products to be implemented in a future waste management configuration, an **efficient and automated sorting is required based on the material composition for defining the respective recycling route and for separating them from non-recyclable parts** (e.g. standard coated parts not including triggers). Due to the lack of recycling options, textiles are often not separately collected or sorted and automotive non-metallic parts shredded as a mixture that is energetically valorised at best, plastics (especially packaging, WEEE) are industrially sorted via spectroscopic identification systems (e.g. NIR, X-ray, colour, laser and combinations)[36] or even hand sorting. The latter is unreliable and *not* suitable for the DECOAT materials as the smart additives cannot be seen by human workers. Partner IRIS has experience in developing tailored fast online VIS/NIR spectroscopic tools for plastic processing monitoring to identify precisely their composition and to detect traces of unwanted substances[37,38].

NIR absorbance spectroscopy is well established as a technique for inline sorting of a wide range of materials, including agricultural products such as food products; minerals[39] and plastic waste[40]. In particular, Steinert GmbH[41] provides a wide range of sorting machines with NIR imaging, handling plastic waste, household waste, compost, building waste, and minerals produced by mining. Waste plastics are especially suited for NIR absorbance characterisation, as they exhibit strong spectral features in the near infra-red. Conversely, most plastics have similar appearance and shapes in the visible wavelength range, so using conventional machine vision for sorting is *not* effective. Allied with this, a large proportion of waste plastic materials will be presented as macroscale objects of a single material type, which means they are easily imaged using the spatial resolution available with common hyperspectral cameras.

Challenges remain in sorting plastics, including distinguishing materials that have multiple component materials, such as coatings, colouring agents and fillers, or separating objects that are dark coloured, or which have been used for containing different materials, including hazardous ones.

Recently, the addition of fluorescent markers has been developed as a way to add specificity to objects that are difficult to sort based on NIR spectroscopy of their basic material type. Such organic dyes absorb light of a particular wavelength and emit light with a longer wavelength. This property, called fluorescence, can be detected with extremely high sensitivity and therefore, only a very small concentration of a fluorescent dye is sufficient. Coatings with a thickness of ~ 1 µm require a dye concentration in the order of some parts per billion, to enable the detection of the fluorescence. Such a small concentration in a thin layer is not visible to the eye and does not affect physical properties like mechanical strength.

In terms of process control of transparent organic coatings, such a system has been recently developed, with the involvement of partner IVV, by the Fraunhofer Polymer Surfaces Alliance POLO®[42]. For the excitation of the dye molecules a (monochromatic) light source (LED, laser) is required, and a digital camera can be used to detect the fluorescence light.

[35] Vinod K. GuptaandTawfik A. Saleh, Synthesis of Carbon Nanotube-Metal Oxides Composites; Adsorption and Photo-degradation
36 Plastic handbook chapter 8: PLASTICS RECYCLING AND BIODEGRADABLE PLASTICS, Susan E. Selke:
37 OPTIJECT project (Grant Agreement no. 262623), www.optiject.eu
38 B2B contracts for the detection of good batches of recyclates as well as for HSI-based monitoring solution for plastic identification in recycling line
39 Steinert KSS Optical Sorting Machine, Steinert GmbH, Widdersdorfer Str. 329 – 331, D-50933 Cologne, Germany. https://steinertglobal.com/mining/industrial-minerals/
40 PICVISA Ecopack, PICVISA Optical Sorting, C/Isaac Newton, 2, 08280 Calaf (Barcelona) España, http://picvisa.com/en/p/productos/ecopack-plastic-sorting-21-18
41 https://steinertglobal.com/magnets-sensor-sorting-units/sensor-sorting/nir-sorting-systems/
42 https://www.polo.fraunhofer.de/de/Aktuelles/downloads.html

Depending on the dye's optical properties additional components such as filters might be necessary. These properties can be used to tag materials and coatings[43] and for example determine coating thickness. In addition, some dyes show a fluorescence effect, whose intensity and wavelength is influenced by the surrounding molecules, the pH, the ionic strength of the medium, or by the viscosity of the matrix, showing a stronger fluorescence depending on the cross-linking degree of the resin.

I**n terms of sorting**, fluorescent markers are already well established as a means of sorting cells or other biological species[44]. A prototype system has been developed to sort small flakes of plastic, by the addition of fluorophores[45], which reduces the minimum size of an object that can be sorted, and handles more complex polymer blends. A commercial fluorescent marker and sorting system for polymers is offered by Polysecure[46]. This system requires the addition of fluorophores to the polymer itself, and in sufficient quantities for detection, and so the amount of fluorescent material required is a limiting factor to the general use of this technology. A new product under development by Nextek Ltd., called PRISM[47], is based on the addition of fluorescent inks to the labels of plastic containers, which strongly reduces the amount of fluorophore required, but presents challenges if the labels have been partially removed or degraded in handling.

Fluorescent markers are typically produced in the form of molecular dyes/ nanoparticles. Dyes are the most cost-effective fluorophore, which makes them attractive for waste sorting. However, where nanoparticles are already deployed as functional additives, such as paramagnetic oscillators, it should be possible to add the functionality of fluorescence to existing particles, or microspheres. Such dual-use particles have been developed for combined cancer-cell imaging and destruction[48]. Commercial fluorescent paramagnetic microspheres are available[49], while customised particles will be investigated that are suitable for material properties of debonding adhesives under development. While carbon nanotubes-based nanocomposites are expected to require fluorescence dyes, other standard metallic nanoparticles (eg iron oxide) can be identified using standard X-ray fluorescence spectroscopy without dye addition. Dispersion of nanoparticles in thin films can be characterised by diffractometric techniques[50].

DECOAT novelty: In terms of process control, fluorescent dyes, applied in ppm (parts per million) concentrations in core and shell of the triggering materials, and resulting in a specific fluorescence will be used in the design the microcapsules. The detection system records the fluorescence intensity distribution, which finally will be converted into volume distribution of the core and shell within the coating. Moreover, the curing states of the core and shell can be imaged by using special dyes. The advantage of such a system is to measure and control all thickness/curing state relevant parameters at the same time during production. In terms of sorting, these fluorescent markers in the core-shell will be unique to the microcapsule and its triggering mechanism, and therefore allow sorting each part to the appropriate channel and process for debonding and separation. The amounts of fluorophore are limited, and localised to the region of the part containing the adhesive layers, so that in the case where more than one additive is present, they can be identified separately, and a multiple stage recycling route can be implemented. All in all, we will research and optimise the **hardware** based on tailored photonic techniques **and software using advanced chemometrics analysis and combining several technologies as required for identifying each relevant material**. They will be adjusted to sort bonded and coated materials. Similar techniques will be used to control proper removal and quantify residues as needed to ensure quality of reprocessed substrate material and for the tracking important parameters of the core shell synthesis, nano/micro composites dispersion and coating thickness as determinants of trigger efficiency.

(iii) Upgraded mechanical and chemical recycling process – mainly related to WP4 work

Trigger assisted mechanical recycling. Mechanical recycling is often most preferred from economic viewpoint. However, it is often limited in capabilities, e.g. for removal of contaminants. A well-known method is using a continuous high speed centrifuge to remove water and other impurities during plastic recycling. To our knowledge, combination with a trigger is not known at pilot or industrial level.

DECOAT novelty: develop **a continuous recycling pilot line to separate the coating from the coated plastics and textiles part by using a high speed industrial centrifuge** (mechanical separation) that **integrates the trigger mechanisms** and that can be **combined with the novel sorting technology**. The idea is to start with grinding in medium size parts (3 x 3 cm), to have the material pass through an activation step (e.g. microwave, IR or steam oven tunnel or chemical bath), to trigger the debonding

43 Holländer et al., (2015). "Process control for thin organic coatings using fluorescence dyes." Progress in Organic Coatings 88: 71-74.
44 David W. Galbraith, *Flow cytometry and fluorescence-activated cell sorting in plants: the past, present, and future,* Biomédica 2010;30(Supl):65-70
45 S. Brunner, P. Fomin, Ch. Kargel, *Automated sorting of polymer flakes: Fluorescence labeling and development of a measurement system prototype,* Waste Management 38 (2015) 49–60
46 Polysecure GmbH, Engesserstrasse 4a, 79108 Freiburg / Germany, http://www.polysecure.eu/en/product-marker-solution/product-marker-solution/
47 Nextek Ltd., Level 2, 1 Kensington Gore, London SW7 2AR, United Kingdom. http://nextek.org/portfolio/sorting-with-markers/
48 Shang-Hsiu Hu and Xiaohu Gao, *Nanocomposites with Spatially Separated Functionalities for Combined Imaging and Magnetolytic Therapy*, J. AM. CHEM. SOC. 2010, 132, 7234–7237
49 Spherotech Fluorescent Magnetic Particles, Spherotech, Inc. 27845 Irma Lee Circle, Unit 101, Lake Forest, IL 60045, USA
50 http://optinanopro.eu/optinanopro01/files/2018/06/ApplicationNote-Texture.pdf

and to start the separation process. The centrifuge movement will permit that the coating, even if its weight is comparable to the weight of the substrate material, will go out of the system fairly easy. Finally, the 'decoated' polymer (plastic parts, textile fibres) will be extruded and pelletized to be used again for melt extrusion. **CO_2 will be introduced during the melt extrusion process to remove residual amounts of coating and VOCs**.

Chemical recycling via CREASOLV process: This is a solvent based selective extraction process developed by partner IVV that allows pure plastics at the molecular level to be recovered due to their specific solubility. It has no by-product (solvents can be used in closed loop endlessly with only few ppm lost in each cycle) and a very low environmental impact demonstrated by peer-reviewed LCAs[51,52]. Unlike general chemical recycling that generate monomers requiring a new polymerization step, Creasolv® generates polymers. One further advantage is that via successive steps with tailored solvents, even mixed input materials can be recycled. Furthermore, due to the mild processing conditions (no supercritical conditions), the polymer maintains its properties as opposed to mechanical recycling. So far, CreaSolv® has been applied to standard thermoplastics like PS, PE, PVC, PA, etc. It has also already been successfully tested on lab-scale for recycling of PVC/PET composites[53].

DECOAT novelty: the **process will be adapted for typical textile and plastics coatings enabling the recovery of selected coating materials**. In combination with DECOAT core/shell microcapsules and lipophilic SAPs, the CreaSolv® treatment will be researched as a potential method of triggering, e.g. by triggering the swelling of SAP's and also as a method to remove remaining contaminants after the application of a previous (mechanical) debonding process.

(iv) Demonstration of circular use and related eco-design– mainly related to WP5 work

Circular use of raw materials of coated parts. DECOAT emphasises on the removal step, i.e. the separation of the coating from the substrate. However, we also want to show that it is possible to recycle the materials, coating and substrate afterwards.

DECOAT novelty: DECOAT ambitions to show this circular use and wants to demonstrate it for one generic coated textile and for one generic coated plastics case. We want to select two such generic systems and will show circularity of the material use. For the actual debonding, we will apply the most suitable technology developed within DECOAT. For the recycling of the coating and of the substrate material, we will rely on the best available technologies at that moment.

Difficult to predict already now but some good candidates: PVC coated PES fabric and ABS coated PUR. The actual recycling of the thermoset PUR is outside the scope of DECOAT but several efforts (eg H2020 projects URBANREC, see also section 2.1.2) are ongoing. DECOAT will rely on these efforts for showing the circular use.

Further, we want to also make prototypes of actual parts for the use cases presented by the end-users, see also next section. These prototypes will be based on eco-design of the coatings, implementing the DECOAT findings. The outcome will be strategies to design single and multilayer coating solutions for textile and plastic substrates using our DECOAT developments.

A public tool will be released that helps to integrate DECOAT technology in the eco-design of coated parts.

1.4.2 Innovation potential

The innovation potential of DECOAT is rooted in the current need for improved high tech material solutions for implementation of cost efficient solutions for recycling of coated textile and plastic parts. A significant contribution can be delivered to the emerging remanufacturing industry in Europe by helping it to extend to plastic parts.

Recycling of coated plastic parts. The current technology for recycling painted parts is based on **chemical bath, which is expensive, prone to contamination, and difficult to implement on small scale** (so for SMEs that need to recycle the scrap). The technology is only available in some large recycling companies and OEMs. This chemical bath has a removal efficiency of ca 95-98,5%. The **current 1,5% of coating that remains causes a dramatic loss of properties in the recycled material**. Taking all this into account, **the innovation potential targeted by DECOAT is to** (i) **find alternatives for the current chemical baths** for a more environmental-friendly process that is easier and cheaper to implement at small scale/companies; (ii) achieve a **removal efficiency of >99% in a single step**, leading to higher quality recycled material at market affordable price.

51 https://polystyreneloop.org/content/8-downloads/presentations/10-lca-of-etics-eol-treatment-of-eps.pdf
52 Freegard, K.; Tan, G.; Morton, R. 3rd interim report on WRAP Project PLA-037. The Waste & Resources Action Programme, Banbury (UK), 2006.
53 U.Knauf et al. Recycling of PVC/PET composites - Production of pure and homogeneous recyclates, and process monitoring – Kunststoffe / Plast Europe Vol 90 (2) 24-26, 2000

Market size coated plastics: European plastic parts production was almost € 37billion euro (Eurostat – 2016 data). Assuming that ca 20% is coated, we find **a European market value for coated plastic parts of more than €7 billion**. For the coating materials themselves (acrylates, polyurethanes,…), the worldwide market on plastics was valued at ca. € 5 billion (2015). The global demand for automotive coatings is ca. 3 million tonnes, with the European share over 20% (2016)[54].

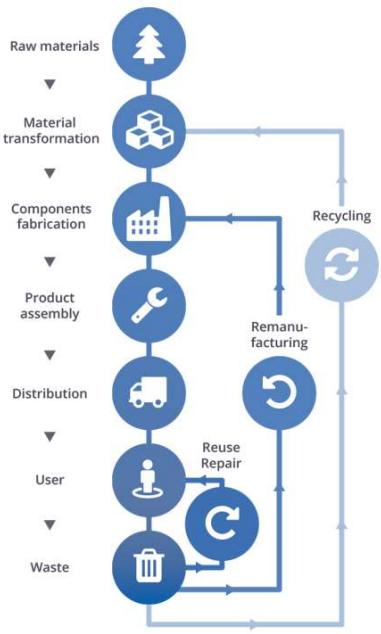
Recycling of coated textiles. In the past recycling of PVC coated PET fabrics and pilot scale recycling has been tried, using chemical recycling, e.g. TexyLoop[55]. Further, textile recycling mainly focusses on shredding of uncoated textile fabrics (typically production waste, also some post-consumer) into short fibres that are reused for non-woven production. And although coated textiles are often used, no economically viable solutions targeting a reasonable value after recycling exist today. The innovation potential targeted by DECOAT is to present coating solutions made with recycling in mind, both from the design and materials side. Moreover, the corresponding techniques will be developed. This innovation potential is on medium-term as **DECOAT will offer a novel option, in-between the existing pure mechanical and chemical recycling, for dealing with coated textiles and plastics**.

Market size coated textiles: **The global coated fabrics market is projected to reach almost €25 billion by 2021**, at a CAGR of more than 4% in terms of value, during the forecast period, 2016 to 2021[56]. The **European textile coatings market is projected to witness growth over 4% up to 2024**. Rising industrial, transportation, and building & construction are driving the market for textile coatings. Along with these, increasing quality and performance standards in textile coating based fabrics are also driving the market. **Recent market potential of microencapsulated products/ additives is estimated at ca. €5 billion, growth ca 5-7%/y**[57] and ca. €1 billion within the technical textile and clothing industry[58], focusing on applications like medical, sportswear and leisure**.**

The global textile coating binders market is dominated by thermoplastics (2015 data): PVC-Polyvinyl Chloride (ca 40%), Polyurethane (ca. 20%) and Acrylics (ca. 7%). PVC is the most used coating for various applications due to its low cost, easy weldability, and easy coloration and printability. PU benefits from being an eco-friendly textile coating, and is widely accepted and preferred by manufacturers in North America and Europe. **Applications for coated textiles are very diverse, but main areas are: transportation**, **building and construction and (technical) clothing**, e.g. for outdoor, workwear, protective clothing[59].

Although currently limited, there is **a strong driver for using coated textiles for daily clothing as it has been shown that the coating is reducing microfibre losses during washing by up to 50%**[60]. But, having to deal with coatings would hamper and delay the current emerging recycling of textiles even more. This is a potentially very large opportunity DECOAT technology can tap into.

Recycling of automotive plastics. During the process of recycling ELV (End-of-Life Vehicles), most of the metal is recovered from cars, and the rest of materials and parts are shredded to obtain what is called ASR (automotive shredder residue)[61]. **This ASR represents 20-25% of the original car weight, it contains typically 20 to almost 50% plastics and most of it is currently landfilled or burned**. The EU directive (2000/53/EC), demands that only 5% of total ELV is landfilled. The plastic separation in the ASR is complex and the purity that is currently feasibly is not enough to have productive mechanical properties in the recycled plastic. It is stated that "***The recovery [of plastic material] has remained marginal****; yet it should develop rapidly now, given the EU requirements regarding material recycling. Manual dismantling allows cleaner recovery, yet it is severely limited by productivity …. Conversely, it seems illusory to recover really clean materials, such as PUR, after passing them through a shredder, even though washing procedures have been tested*"[61]. **Therefore, rather than focussing on small incremental improvements, DECOAT proposes a radical novel approach.**



The **global automotive industry is set to reach 114 million cars in worldwide sales annually by 2024**, with an expected growth on average of 2.8%[62]. Within the EC report 'GEAR 2030'[63] (2017), the opportunity for building novel or extending existing value chains

54 https://www.zionmarketresearch.com/news/global-plastic-coating-market and http://www.european-coatings.com/Markets-corr
55 Recycling of flexible composite materials – see: www.texyloop.com
56 https://www.marketsandmarkets.com/PressReleases/coated-fabric.asp and https://globenewswire.com/news-release/2016/11 Market-2016-2021-Market-is-estimated-to-reach-USD-20-41-billion-in-2016-and-is-projected-to-reach-USD-24-91-billior
57 Source: Gerald Rafler, Head of the Fraunhofer Institut Applied Polymer Science, Potsdam (2016)
58 Estimate based on European Textile Association statistical data on technical textiles and clothing segments
59 https://www.marketsandmarkets.com/PressReleases/textile-coatings.asp (2016)
60 A New Textiles Economy (2017), see: https://www.ellenmacarthurfoundation.org/publications/a-new-textiles-economy-redesigning-fashions-future
61 A. Buekens, X. Zhou, Recycling plastics from automotive shredder residues: a review, J Mater Cycles Waste Manag (2014) 16:398–414
62 https://www.consultancy.uk/news/13900/global-car-market-to-break-through-100-million-in-sales-by-2020
63 https://ec.europa.eu/docsroom/documents/26081/attachments/1/translations/en/renditions/native

for a circular automotive sector is highlighted, as well the possibility for novel business models, eg for the recycling or re-use of (spare) parts. DECOAT feeds directly into this opportunity, by providing novel technological solutions for both disassembly and recycling. The **high-end premium car segment which uses a lot of aesthetics coatings** (ca. 13% of MAIER's turnover) **is 'booming'**: it increased with more than 50% from 2010 to 2016[64].

Of special interest for the automotive sector is **the increased use of composite parts, especially in electric vehicles**. This currently hampers the EoL processing as recycling of composite parts is much more difficult as (coated) metal parts. Therefore vehicles are designed that are easier to recycle, eg by making them easier to (partially) disassemble, even in a robotised manner. This offers a huge opportunity for the recycling industry and linked activities. DECOAT technology for recycling parts fits perfectly in this scheme.

Opportunities exist also for trucks and busses, the latter are covered by partner MBT. Although less common then cars, there are roughly 1 million so-called HDOR (Heavy duty and off-road vehicles) discarded every year. Disassembly is much more common and far advanced with some brands for metal parts (iron, steel, aluminium, up to ca 75-85% of the total weight). **Remaining challenge is dealing with complex multi-material parts** (with electronics). DECOAT technology could contribute.

Emerging market: recycling of smart textiles and plastics with integrated electronics. **The market for smart textiles and wearables is expected to enter an exponential growth phase** from an expected total world market of nearly 1.5 bn US$ by 2021[65] to almost 3 bn US$ for the EU market alone in 2024[66]. Eg **DECOAT partners MBT, MAIER and CTAG are working on integrating electronics into various (interior) car elements**. Next to the actual manufacturing, a missing solution to recycle the smart plastics hampers the market introduction. Therefore, we will also look into this emerging automotive application.

Remanufacturing is the process of returning a used product to its original performance specifications (from the customers' view). It is a strategy to reduce manufacturing's environmental impacts while boosting profitability. If considered during the early stages of a product's design, the concept of 'active disassembly' can be applied enabling the rapid and non-destructive disassembly of products with minimal component damage, which facilitates remanufacturing. DECOAT technology can aid to implement active disassembly, easing the actual disassembly (DECOAT – 1st level) and for making it easier to upgrade used parts by removing the coating so that it can be recoated. **Remanufacturing is especially of interest for the automotive market**. The worldwide automotive parts remanufacturing market (focus on metal parts) was valued at US$ 33.16 Billion in 2016 and is estimated **to grow at a CAGR of 6.6% from 2017 to 2025**[67]. DECOAT technology could help to extend it also to plastics parts. Note that this is much stronger than the actual automotive market (worldwide 2.8%, in EU only 1.7%). Economic benefits for both remanufacturing companies as well as the customers are considered as the biggest driver for the overall automotive parts remanufacturing market. Remanufacturing aids in reducing cost pertaining to major parameters such as material, machining, production time, energy, fuel and labour costs, among other. In addition, present-day automobiles are made flexible so that they can easily incorporate remanufactured parts without affecting the performance of the vehicle.

Remanufacturing goes wider than the automotive sector: the EU Remanufacturing Network estimates the current size of the **EU remanufacturing industry to be almost €30bn, employing around 190,000 people**. Their market study shows that **the industry could triple to €90bn by 2030** and employ as many as 255,000 people[68]. **As DECOAT technology will also be able to apply on large full parts, it can contribute to and benefit from the remanufacturing opportunity**. The ERN study[69] identified the top barriers for remanufacturing becoming more widespread, i.e. volume/availability of a 'core' (used part intended to become a remanufactured product) and quality of the core. For coated parts, DECOAT technology answers to those barriers. The aerospace sector was shown as the sector where remanufacturing is currently being used most. On 2nd and 3rd place are automotive and EEE (Electrical and Electronic Equipment). The INDAR technology of partner RESCOLL is indeed already emerging in the aerospace sector. Our goal is to make it also enter other sectors, like automotive and EEE. This shows that DECOAT will also be useful for further development for disassembly of parts for these sectors (and debonding of coatings).

The global adhesives market was valued at ca. €35billion in 2015 and is projected to reach ca. €50billion by 2021[70]. The **share of debondable adhesives is nowadays only ca. 0.5% of the general adhesives market**. The water-borne adhesives (= DECOAT scope) account for ca 50% of the global market. If DECOAT results reach 1% of this market, we can address a global market of ca. €170-250million.

The above motivates the selection of the DECOAT use cases: watertight backpack for outdoor gear (VAUDE), power switches and sockets for household electronics (PANASONICS) and automotive parts (with integrated electronics) (MBT,

64 Statistica Charts: https://www.statista.com/chart/9309/super-luxury-car-sales-are-booming/
65 Kamitis – Smart Textiles Study 2016-2021
66 Grand View Research - Smart Textiles Market, Smart Fabrics Industry Report (2017)
67 https://www.researchandmarkets.com/research/9b2rfv/global_automotive?w=4 (2017)
68https://www.circle-economy.com/european-remanufacturing-industry-estimated-at-e30bn-with-potential-to-triple-by-2030/#.Wy-ubExuJZU (2016)
69 Remanufacturing Market Study by ERN, Oakdene Hollins (2015)
70 Global Adhesives and Sealants Market– Trends and Forecasts (2016 - 2021), edited in October 2016 by MORDOR INTELLIGENCE

MAIER) as applications and sectors where recycling of coated materials is needed but no economically viable solutions are available yet.

FREEDOM TO OPERATE. The consortium carried out a patent survey to specify possible opportunities not only to achieve knowledge advancements, but also to proceed in technological advancements in terms of patenting. The survey was conducted in the spirit of investigating and detecting unexplored areas and approaches of the field and to avoid possible overlapping with existed protected intellectual knowledge. The consortium decided that although a number of patents exist in similar fields, DECOAT project offers a huge opportunity to achieve technological advancements beyond the state of the art.

Patent No – Title	Short Description	DECOAT's Innovation potential
CN105710044A: Intelligent classification system for recoverable garbage	Intelligent classification for recoverable garbage: infrared (IR) sensor, metal detector, core control classification module and voice prompt module (IR sensor and metal detector used for detecting type of garbage.	**DECOAT** will develop an automated sorting technology based on triggerable additives.
CN100386453C: Method for decoating on waste plastic metal	The present invention relates to a method of recycling scrap metal, in particular a plastic waste containing metal plating layer simultaneously recycling waste plastic and metal plating method faded.	**DECOAT** will develop a method for decoating a paint layer from a plastic substrate in order to recycle both the coating and the substrate
US20140287299A1: Heat-Debonding Adhesives	A debonding tool provides current to conductive contacts on the conductive layer for generating heat in the heat-generating layer when it is desired to debond the structures that are attached together in electronic devices, using the adhesive member.	**DECOAT** will develop triggerable primers for interfacial debonding for electronic devices, automotive parts and textiles
US5100494A: Structural bonding and debonding system	The pieces are debonded or separated by passing a relatively high electrical power through the heating element, to soften or melt the adhesive.	**DECOAT** will use triggers such as microwave, Rf irradiation, heat and steam together with smart nano-materials, for softening/melting adhesives.
US7901532B2: Method and apparatus for bonding and debonding adhesive interface surfaces	Invention utilises thermoexpanadable microspheres and thermal energy to debond interfaces in an adhesive system or as vehicle carriers. It discloses a method of curing the adhesive system prior to the debonding step so that the same adhesive system may be used for both phases. It is especially useful in the automotive industry for end of vehicle life dismantling.	**DECOAT** will use super absorbent polymers as microspheres which will react to moisture and heat. The applications are not only limited to automotive industry, but also to electronics and textiles
US20150202858A1: Laser film debonding method	A laser-based coating removal method debonds a film from a substrate rather than ablating the film. Laser light is transmitted through a transparent film to an underlying bonding layer for bonding the film to one or more additional films and/or a substrate. The laser light is absorbed at the bonding layer and the transparent film is released. In some embodiments, after the transparent film is released it is able to be physically removed.	**DECOAT** will use microwave and Rf irradiation instead of laser light. Automated optical sorting will be also implemented.
US6288158B1: Modified superabsorbent polymer based on polyacrylonitrile emulsions	The surface of a polymer containing hydrolyzed polyacrylonitrile in the form of particulates, is modified by crosslinking in a water/alcohol reaction mixture and in the presence of silica. The resulting product features improved absorbency under load and lesser degree of gel blocking	**DECOAT** will synthesize a new generation of SAP's functionalising their surface with thermo-responsive coatings and decorating them with metal oxide nanoparticles.
US6537341B2: Process for recycling fiber composite materials	A process for recycling composite materials that comprise fibers and a matrix includes exposing the composite material to electromagnetic waves in the form of microwaves. The electromagnetic waves induce an introduction of energy into	**DECOAT** will develop a recycling route based on microwaves and Rf irradiation for coated plastics and textiles. In combination with the above, the

	the composite material, leading to a separation of fibers and matrix. The separation operation can be assisted by a suitable solvent.	Creasolv process will be act synergistically.

In spite of the existing prior art (described in section 1.4.1), according to the knowledge of the partners, no patents exist that block the envisioned developments and innovations. Quite on the contrary, several **DECOAT partners own key patents related to the developments. An overview of the most important ones:**

DECOAT scope	**DECOAT partner – relevant patents**
Debonding adhesives	**RESCOLL**: proprietary INDAR technology via patents EP2519596 (B1), EP2519565 (A1), EP2265681 (B1), EP1814935 (B1), EP1611217 (B1)
Microcapsule production & application	**DEVAN**: has an extensive know-how, through the PCT/IB2006/050605 patent, for the grafting of a covalent bond between the microcapsules' shells and the textile fibres. The incorporation of these microcapsules into fabric substrates creates new functionalities for the treated materials.
Microwaveable trigger (for yarns)	**CTECH**: 'Article assembly disassembly system' - Patent WO2013/189956A1
Nano- and microcapsules	**NTUA**: "Multifunctional Coatings including Nanocontainers with Self-Healing, Antibacterial and Photocatalytic Properties" - Patent no 1007721; 'Hybrid Coatings including Nanocontainers with Antifouling Properties" - Patent no 1007729; "Functional Multi-stimuli responsive polymeric nanocontainers-microcontainers as drug delivery systems" - Patent no 1007882.
Creasolv®	**IVV**: "Process for recycling soluble polymers or polymer blends from plastic containing materials" - Patent No EP0894818B1; "Method for increasing the concentration of at least one polymer from a polymer containing waste material, and polymer recyclate" - DE102013212813A1; "Method for recycling plastics and use of thereof" - EP1888677B1.

The DECOAT consortium is convinced that the triggered debonding approach presented here is still largely uncharted and that opportunities are multiple to patent IP, including options to extend and elaborate some of the existing patents of DECOAT partners.

2 Impact

'Do not go where the path may lead, go instead where there is no path and leave a trail'

(Ralph Waldo Emerson, writer)

2.1 Expected impacts

DECOAT fits extremely well in the European Circular Economy strategy as it aims for a systemic innovation for the recycling of coated textile and plastic materials. The materials solutions will build further on latest (carbon) nanotechnology, microencapsulation and polymer developments and lead to several scientific and technological breakthroughs. DECOAT results will create new markets for triggerable additives, optical material detection systems and for recycling equipment and processes. Together this amounts to over 150 million annual turnover, i.e. the potential to directly create ca. 500 jobs. Moreover, DECOAT materials and process innovations will give a competitive edge to a relevant share of the European companies active along the coated textiles and plastics value chains that could generate a total turnover of ca. €1.7 billion, implying that DECOAT would further help to secure ca 5.000 jobs.

2.1.1 Expected impacts set out in the work programme

In the subsections below, we list how DECOAT will respond to each of the expected impacts of the call. We focus on the impact within and beyond DECOAT and try to illustrate both aspects.

2.1.1.1 Meeting the EU's circular economy and environmental targets while demonstrating a clear benefit, i.e. more efficient or economic than the state of the art in order to enable market uptake in the short to medium term

Efficiency aspect: The current disposal methods for coated textiles and plastics are landfilling, incineration or (very limitedly) chemical bath recycling, that all have severe disadvantages. Incineration[71] conventionally leads to the release of significant greenhouse gas CO_2, pollutants like NO_x, SO_2, and particulate matter. Additionally, heavy metals and dioxins are emitted. Given the high energy content, much of the heat is recovered as energy source. However, the residual product consists of bottom ash, fly ash, and air pollution control residue, which is currently also landfilled. The incineration process involves a flue gas cleaning process, which may give rise to externalities to wastewater. For **landfill**[71] disposal, the main emissions are CO_2 and CH_4. Moreover, leachate is released to the surrounding soil and water. In addition to environmental impact, both incineration and landfill processes require facilities with serious disadvantages, e.g. for the people living in the vicinity. **Chemical baths** are currently used to remove coatings and to recycle the left-over fraction but they are very limited and cumbersome from environmental side (see section 1.4.2). **These dramatic impacts will be avoided by the DECOAT innovations**, as instead of being disposed of, the majority of the original material can get a secondary life-cycle. This saves on the production of virgin material, thus contributing to significantly reducing emissions throughout the value chain.

DECOAT offers a solution to coated plastics and textiles recycling via the development of triggered debonding of coatings via improved mechanical or chemical recycling, resulting in **a highly promising, efficient alternative to current recycling and material design standards**.

Economic aspect: for doing the extra sorting and activation of the trigger, an additional cost during recycling of ca €80-100/ton is estimated (including OPEX and CAPEX), largely depending on the stimulation mechanism (e.g. infrared vs microwave). This is an economic attractive proposition as the current process of non-coated plastics recycling, *including extrusion* and pre-treatment (grinding, washing, drying) costs ca. €300-400/ton. So, **coated parts recycling can be enabled by ca. 20% extra cost compared to non-coated parts, which is close to but below the only current solution** (solvolysis ~ €500/ton).

Contribution to EU's circular economy and environmental targets, which specifically recommends accounting for the entire plastics value chain[6]. The primary focus of the EU circular economy for plastic use and production is to promote the reuse, repair, and recycling needs of plastic value chains through enhanced sustainable design. DECOAT directly addresses these challenges by developing a smarter initial design to plastic and textile coating that can be removed upon end-of-life status, allowing a high percentage of the original material to go on to a second life-cycle. This directly supports the EU's goals to achieve a "modern, low-carbon, resource and energy-efficient economy" while making "a tangible contribution to reaching the 2030 Sustainable Development Goals and the Paris Agreement".

2.1.1.2 Create new technologies and business opportunities for the recycling industry across Europe, especially in the area of composites and plastics where the challenge is high

Key DECOAT result: **novel technologies for triggerable coatings** (←KO1, KO2)

71 http://ec.europa.eu/environment/waste/studies/pdf/econ_eva_landfill_report.pdf

For: (i) R&D driven SMEs in area of chemistry, formulation, paint, adhesives, … (ii) product designers & developers

Impact: For group (i): **a novel market will be created of (additives for) coatings, paints and adhesives that are fulfilling the recycling need**. The global adhesives market was valued at ca. €44billion in 2016, and is expected to register a CAGR of 5% during the forecast period (2018-2023)[72]. **This sector contributes more than €13 billion to the EU economy and employs more than 41,000 people**. **The share of debondable adhesives is nowadays only ca. 0.5% or ca. €65 million of the general adhesives market**. Our results will contribute to increase this market share as we expect to **generate a market of DECOAT triggerable additives of ca. €20 million** (see part D2 of the PEDR). The DECOAT partners will be ideally positioned to capitalise on this growing market: **RESCOLL** expects the project to lay the basis for a new product line of debondable adhesives and primers for textiles and plastics products. RESCOLL has the capacity to scale-up to production via its subsidiary company STILZ Chimie to which it wants to transfer the developed technology (royalty based agreement), and which is currently capable of producing up to 20 tons/year. **DEVAN** will be able to continue to fuel the growth of their turnover in microcapsules products based on DECOAT results. **CTECH** is using already microwaveable additives but so far only in yarns via the extrusion process. DECOAT will allow different market sectors to be entered, notably for coatings for textile and plastics.

For group (ii): the DECOAT concepts will be useful for designers and product developers as they can use these new concepts for making **'recyclable-by-design' products**. Further, DECOAT will **boost repair services** as DECOAT results can also be used for developing products that allow easy repair by enabling easy exchange of parts (via another use of the disassembly concept of DECOAT). This is relevant for local workshops that want to offer repair services but also for a large brand like **VAUDE who already offers repair services**. Finally, **DECOAT triggerable material solutions will also be useful for other application areas** where it is needed to separate materials for recycling. This is for example the case for coated plastic parts in **construction or for the composites sector,** fueling also additional sectors like aeronautics and marine etc. For the latter, DECOAT solutions could be used to weaken the interface matrix-reinforcing fibres so that recycling becomes easier. Further, as mentioned before DECOAT solutions are also an **opportunity to recycle the emerging smart textiles and plastics and wearables**. ALL DECOAT partners will benefit from this exploitation beyond the initial DECOAT scope but it will be especially interesting for the research partners in the project (CTB, AIMPLAS, IVV, CTAG and NTUA) for whom this offers a way to exploit results beyond DECOAT.

Key DECOAT result: **novel recycling processes** based on combination of mechanical and chemical recycling with a triggerable material (←KO4) and the linked automated sorting (←KO3).

For: (SME) players in the recycling industry, that are more and more emerging; established recyclers of plastics and companies that design and build turn-key recycling lines

Impact: typically for common coated materials, only mechanical recycling is now economically feasible as full chemical recycling is *not* cost efficient. **DECOAT (via partners IRIS and AIMPLAS) offers a way to upgrade existing mechanical processes** by adding tools that requires limited investment, **for activating the trigger and for automated sorting**. Once DECOAT triggerable materials come on the market **existing recycling lines will have to be upgraded for the sorting**.

Key DECOAT result: **demonstration of circular use of the demo cases products** (←KO5).

For: (i) DECOAT end-users PANASONIC, MBT and VAUDE; (ii) manufacturers, and their suppliers, of coated textile and plastics based consumer goods, eg TIER 1 for automotive, textile manufacturing sector, coatings and paints industry, …

Impact: For group (i) DECOAT will create direct business opportunities. **Partner MAIER employs ca. 2400 people with a 300M€ turnover, and owns plants in Spain, Italy, the UK and Czech Republic**. It is supplier of plastic components for the most important automotive OEMs (e.g. PSA Group, TOYOTA, AUDI, RENAULT, JAGUAR-Land Rover, GM, BMW, NISSAN, etc.). Being the first on the automotive market to offer DECOAT triggerable coating solutions is expected to give an important competitive edge that could relevantly boost ca. 20-25% of the product portfolio. **Partner PANASONIC** is part of one of the largest brands all over the world and the Panasonic Eco Solutions Türkiye plant is one of the largest companies that produce on this scale in one place in Europe (100.000.000 pcs/year). PANASONIC employs more than 250.000 people all over the world. Panasonic Eco Solutions Türkiye focusses on the products that are mainly targeted in DECOAT, e.g. Power Switches and Sockets, Group-Sockets, Fuse Boxes, Low-Voltage Switchgear Products, Meters, Automated Meter Reading Systems, Building Automation Systems. DECOAT results are relevant for 15% of the product portfolio. **Partner VAUDE** ranks among Europe's leading outdoor brand names and employs over 500 people in Germany. Its product portfolio comprises full range of outdoor gear: backpacks, shoes, tents, garments. The use of more sustainable and recyclable materials opens up commercial opportunities as being at the forefront there is a winning marketing argument. Goal is to integrate the technology as a start in a 'coated backpack' line, comprising ca 10.000 backpacks per year (average weight ca 1.5kg). This would be ideal to reach the larger public. **Partner MBT**, ca. 6000 employees, is one of the largest manufacturing companies in Turkey. Mercedes-Benz Türk is the country's leading bus and truck exporter. Today 80 % of the buses produced by Mercedes-Benz Türk are exported to more

72 Innovations in Multi-material Joining Technologies, Frost & Sullivan 2013

than 70 countries, especially to Western Europe. DECOAT will bring MBT the important competitive and commercial advantage of recycling know-how and possibility for procurement of cheaper (secondary) raw materials. MBT has turnover of ca. €3.3 billion, the share of the busses in it is ca. €520million, the share of the trucks is ca. €870million.

For group (ii), the **wider European community of manufacturers and users of coated plastics and textiles, DECOAT will offer concrete examples that can be adopted to vastly extend the share of products (parts) that can be recycled in their own product portfolio**. This will leverage the DECOAT impact as this wider community is large in EU: total plastics industry: ca €350 billion turnover (data from EUPC); textile & clothing sector: ca €170 billion turnover (data from EURATEX).

Key DECOAT result: **environmental, safety & economic assessment** of triggerable material solutions (←KO6, KO7).

For: manufacturers faced with need to separate complex products

Impact: ECOMAT, IRES and OSM will be able to use the gained expertise for offering consultancy services. Using the DECOAT tool (see Task 6.5), industrial partners will be able to channel the impact assessment of the products to their life cycle scenario and respective business models.

DECOAT business opportunities and market creation potential will lead to job creation: DECOAT results are relevant for ca. 3.5 million tons of waste worldwide (see section 2.1.1.3). As argued in part D2 of the PEDR, DECOAT is well-geared to reach ca. 280kton (8% share, which will bring important opportunities all along the value chain. For calculating the opportunities we refer to part D2 and we assume that 1 full time job corresponds to €350k. This leads to the opportunity of installing the DECOAT recycling equipment (optical detection, MW equipment, Creasolv system), which is estimated in total at €35million. Although this is a one-time opportunity, the related post-installation services, maintenance and upgrades are expected at 20% of the initial installation cost, leading to a continuing turnover of ca €7million annually or ca 20 jobs. Further we have the market of triggerable additives: €20 million per year or ca. 60 jobs; the recycling of the coated parts: €140million per year or ca. 400 jobs. **In total this means DECOAT has the potential to directly create ca. 500 direct jobs.**

DECOAT developments will contribute to the competitiveness of the European manufacturing industry, which is an important employer, for example: automotive sector: ca. 3 million manufacturing jobs (EC data); textile & clothing sector: 1.7 million jobs (EURATEX data). It is calculated in part D2 of the PEDR that **DECOAT materials and process innovations will give a competitive edge to European companies active all along the coated textiles and plastics value chains that is relevant to a total turnover of ca. €1.7 billion, implying that DECOAT will help to secure ca 5.000 jobs.**

2.1.1.3 Demonstrate a potential reduction in landfill waste volume by > 50%

DECOAT will lead to a decrease in landfilling by an estimated 75% as the largest fraction of coated parts (the bulk) will be re-used. Indeed, the coating itself does (in most applications) not relevantly participate to the volume. Nevertheless, to account for potential damage at the coating interface, we assume a 5% loss of the bulk volume. Depending on the type of recycling used in combination with the trigger, it might be possible to recycle the coating material. However, here we assume the coating itself is *not* recuperated, so 5% volume remains landfilled. One further element to take into account is that the recycling process of the bulk itself will *not* be perfect. A fair estimate is to assume 80% efficiency. So, ca. 75% of the volume of the initial coated waste can be recycled, see also the concrete case study in section 2.1.4.

This reduction is relevant for in total almost 3.5 million tons of coated plastic from cars, electrical and electronic equipment and coated textiles. This number is calculated based on the following data and estimates[73]:

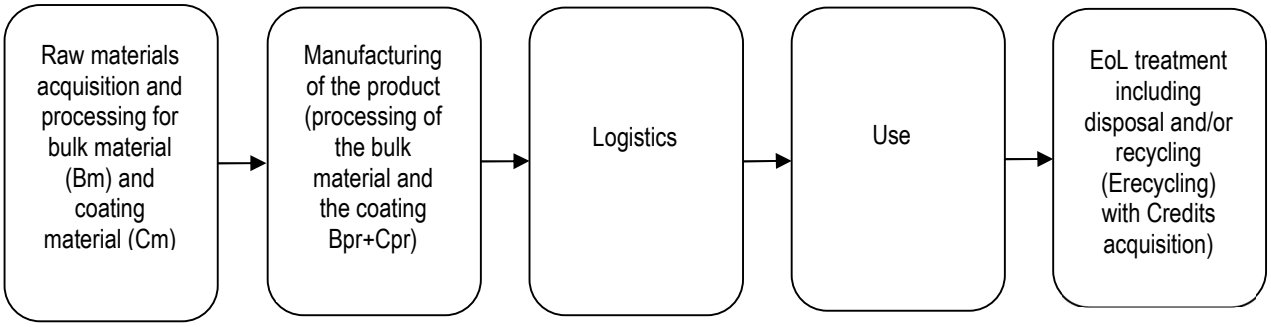
- **Automotive parts: ca. 2.1 million tons of coated plastic come from end-of-life vehicles (ELV) per year**. Motivation: 27 million cars are discarded every year[74], at 225 kg plastic per car (15% of average total car weight[75]) leads to ca. 6 million tons plastics ELV from cars of which ca. 35% is coated/ painted. As these coated plastics are part of the ASR of EoL vehicles, **this fraction is mainly landfilled** (see also section 1.4.2).
- **Heavy vehicles parts**: **ca. 800 ktons** of coated plastics. Motivation: ca. 1 million heavy duty vehicles reach yearly the EoL stage in Europe[76]. Taking the average weight per vehicle (15 ton[77]) and 15% of plastics we have ca. 2250ktons plastic waste. We assume again that ca. 35% of these parts is coated/ painted.
- **Electronic goods: ca 200 ktons of painted plastic come from electronic goods ((W)EEE = (waste) electrical and electronic equipment) per year.** Motivation: ca. 3.45 million ton of WEEE is yearly collected (Eurostat 2015). On average, ca 20% is plastics[78]. Further, we assume that ca. 30% is coated/painted, which leads to ca 200ktons.
- **Coated textiles: ca 350 ktons are wasted per year** that could be recycled using DECOAT solutions. Motivation: the total

73 The estimate of the 'coated' fraction for the various sources is based on the best available knowledge of the (end-user) partners of the DECOAT consortium
74 Automotive Recycling. Devalued is now revalued' Plastic Market watch. SPI (2016)
75 'Automotive The world moves with plastics". Plastic Europe (2013)
76 M. Saidani et al. 'Heavy vehicles on the road towards the circular economy: Analysis and comparison with the automotive industry' Vol 135 108-122 (2018).
77 'Research on the Weight of Buses and Touring Coaches', see: https://www.iru.org/sites/default/files/2016-01/en-nea-bus-2007.pdf
78 Seyring et al. (2015). Study on WEEE recovery targets, preparation for re-use targets and method for calculation of the recovery targets

market for coated textiles is ca. 1 million ton / year[79]. The combined share of PU, acrylic and PVC (=DECOAT scope) is ca 70%. Furthermore, due to improved collection practices, we estimate that ca 50% of the products will be available for recycling (currently only 25%[80] but increasing trend), which leads to ca. 350 ktons.

2.1.1.4 Reduction of the carbon footprint of the corresponding products by > 30% (based on a full Life Cycle Assessment).

The actual carbon footprint, and other relevant environmental parameters, will be calculated during DECOAT via a Life Cycle Assessment (LCA). The study will include all life cycle stages as represented in the figure below, impacts of each of them shall be considered during the assessment. At this moment, we estimate the potential of the proposed technologies for carbon footprint reduction. To do so**, we consider the first use of virgin materials in an open loop product system, where no changes occur in the inherent properties of the recycled material. Here, the "business as usual" baseline scenario comprises the assumed current End-of-Life (EoL) treatment (landfill and incineration), and the DECOAT scenario comprises recycling of the bulk material from the products in their EoL**. This recycling would allow to receive the "emission credits" due to avoided emissions during the virgin bulk material production that the recycled material is intended to replace (avoided burden approach). These credits are deducted from the total impact of the product as a part of the EoL stage. In exchange, the impacts of the triggerable debonding process and the average composition of the coated parts to be recycled are also allocated to the product system.



The emissions from logistics, use, andfill disposal and incineration are assumed to be equal for both EoL scenarios, and thus not included in the inventory for the current calculations. This and all other methodological choices will be further analysed and evaluated in dedicated sensitivity analyses during the execution of the full LCA.
Thus, the total impact can be expressed as follows:

$$\mathbf{TOTAL = Bm + Cm + Bpr + Cpr + EoL, \text{ where}}$$
$$\mathbf{EoL_{avoidedburden} = E_{recycling} - Credits}$$

However, the credits shall be calculated based on the impact of the primary production of bulk material and recycling efficiency. **The formula was used to derive the generalized calculations on the potential of DECOAT to reduce environmental impacts of product systems, as well as a case study of a coated PVC product was made to validate the general approach and make it tangible**. Tables below show the results, including further assumptions for the generalized calculations.

Baseline scenario:

Intermediate	Calculation	Generalised: CO2eq		Example PVC			
				waste		CO2eq	
		calculation	assumptions	value	unit	value	unit
Virgin bulk material	Bm	Bm = 0.8*B	Bm comprises 80% of the material impact	0,95	kg/kg	1,9	kgCO2/kg
Material processing	Bpr	Bpr = 0.2*B	Bpr comprises 20% of the material impact	0	kg/kg	0,38	kgCO2/kg
Coating material	Cm	Cm = 0.5*C	Cm comprises 50% of the coating impact	0,05	kg/kg	0,2	kgCO2/kg
Coating processing	Cpr	Cpr = 0.5*C	Cpr comprises 50% of the coating impact	0	kg/kg	0,25	kgCO2/kg
Total Baseline	T = Bm+Bpr+Cm+Cpr	T=B+C		1	kg/kg	**2,73**	**kgCO2/kg**

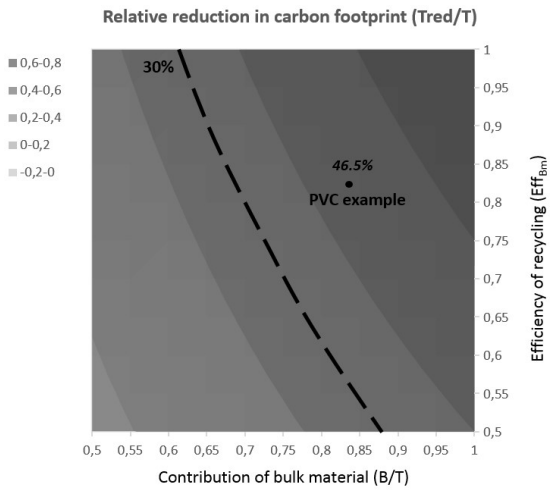
79 MarketsAndMarkets report (2017)
80 http://www.foeeurope.org

DECOAT scenario:

Intermediate	Calculation	Generalised: CO2eq		Example PVC			
				waste		CO2eq	
		calculation	assumptions	value	unit	value	unit
Virgin bulk material	Bm	Bm = 0.8*B		0,95	kg/kg	1,9	kgCO2/kg
Material processing	Bpr	Bpr = 0.2*B		0	kg/kg	0,38	kgCO2/kg
Coating material	Cm	Cm = 0.5*C		0,05	kg/kg	0,2	kgCO2/kg
Coating processing	Cpr	Cpr = 0.5*C		0	kg/kg	0,25	kgCO2/kg
DECOAT debonding ($E_{recycling}$)	R	R = Cpr = 0.5*C		0	kg/kg	0,25	kgCO2/kg
Bulk material recovering efficiency	EffBm	EffBm		80	%	80	%
Coating material recovering efficiency	EffCm	EffCm = 0	Pessimistic assumption, no coating material recovery	0	%	0	%
Recovering bulk material (Credits)	Bmrec = Bm*EffBm	Bmrec = 0.8*B*EffBm		0,76	kg/kg	1,52	kgCO2/kg
Recovering coating material (Credits)	Cmrec = Cm*EffCm	Cmrec = 0		0	kg/kg	0	kgCO2/kg
Total DECOAT	Trec = Bm+Bpr+Cm+Cpr + R – Bmrec - Cmrec	Trec = B+1.5*C - EffBm*0,8*B		0,24	kg/kg	**1,46**	**kgCO2/kg**

Environmental impact reduction in two scenarios:

Intermediate	Calculation	Generalised: CO2eq		Example PVC			
				waste		CO2eq	
		calculation	assumption	value	unit	value	unit
Total Baseline	T	T=B+C		1	kg/kg	2,73	kgCO2/kg
Total DECOAT	Trec	Trec = B+1.5*C - EffBm*0,8*B		0,24	kg/kg	1,46	kgCO2/kg
Footprint reduction (ΔEOL)	Tred = Trec – T	Tred = 0.5*C - 0.8*B*EffBm		-0,76	kg/kg	-1,27	kgCO2/kg
DECOAT impact (relative reduction)	**Trel=-Tred/T**	**Trel=EffB*80%(B/T) – 50%(1-B/T)**		**76,0**	**%**	**46,5**	**%**



On the generalized level, the relative reduction of environmental impacts due to DECOAT introduction can be expressed as a function of the bulk material recovery efficiency (recycling rate), and a contribution of bulk material to the total product's environmental impact. The distribution of relative reduction values dependent on the two parameters is presented in the graph on the right.

With the assumptions made about the contribution of the bulk material into the total environmental impact of the product (80%) it is necessary then to keep the recycling rate (efficiency of recycling) above 62.5%. However, for the target applications, the B/T ratio if not expected to in the lower ranges, and the recycling rate is expected to be designed above 95%. Thus, the products' performance will realistically always be in area on right upper side of dashed line, i.e. **we will comply with and even outperform this expected impact.** Additional benefits that were not included would further reduce the environmental impact during waste handling i.e. the reduced emissions that result from not incinerating or landfilling the materials, and possible recycling of the coating material.

Under current circumstances, the production of 1kg of coated PVC results in 2.73 kgCO2/kg virgin material. **The DECOAT process would cut the carbon emission by 46%** (1.46 kgCO2/kg virgin material), assuming that the DECOAT process has a material recovery efficiency Eff_B of 80% (this reflects a 20% loss due to processing energy and lost base material during processing). This also means **the DECOAT process would result in a 76% reduction in waste generation** (80% efficiency results in 0.76 kg virgin material recovered per kg virgin material input into the system). **These initial outcomes of the case**

study show that DECOAT expected impacts exceed the desired 30% reduction in carbon footprint and also the 50% reduction of landfilling (see section 2.1.3).

2.1.2 Improving innovation capacity and market creating potential

Innovation capacity comprises the internal ability to respond to external opportunities with an innovative solution that leads to profitable business. **DECOAT claims to significantly increase the innovation capacity in the coatings, adhesives and paintings sector and the related value chain (material suppliers, end users) by combining two external opportunities: (i) the global need for recyclable coated parts (= market pull) and (ii) the large(r) scale and low(er) cost availability of smart multifunctional additives such as nano- and microparticles (carbon-based among others) (= technology push).** DECOAT developments and results aim at capitalising these challenges, both from pure technical standpoint as from supporting activities (standardisation, LCA, benchmarking) as described below.

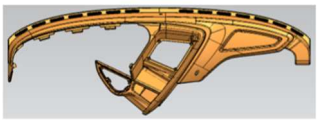
DECOAT will boost the innovation capacity of the debonding technology providing partners (RESCOLL, DEVAN, CTECH, IRIS, MAIER) by demonstrating the advantages of their materials and processes for different use cases for VAUDE, PANASONIC, MBT and MAIER. A direct advantage for the end-users will be the reduced cost for sourcing material as the cost of recycled material (at ca. €500/ton, see section 2.1.1.1) will be lower as virgin ones.

An important aspect for the improved innovation capacity for these technology providers, mainly SMEs, is that the **DECOAT partners with high visibility and brand ownership (VAUDE, PANASONIC and MBT) will support these technology providers** in getting the DECOAT solutions into the market. Such flagship early adopters are important to realise the innovation potential (e.g. for market pull effect and 'enforcing' their suppliers to use DECOAT technology, guarantee for re-use of recycled materials, marketing efforts) from which DECOAT will form the basis. The end use cases are summarised in the table below.

The goal is that by the end of DECOAT **exploitation agreements will be signed among DECOAT partners for the further development of the results.** Given the expectation that full market readiness of the DECOAT developments will still take some extra investments, this will be important to set the follow-up actions to get to a higher TRL level towards a successful market deployment of DECOAT results on medium to long term.

DECOAT use cases			
	Use case	**Recycling drive & challenge**	**Estimated impact**
VAUDE Water-tight bike bag 	Fabric material: PES, TPU coating Ca 28%-72% substrate-coating(!) Weight fabric: ca 700g/m2 Currently no solution to recycle these coated fabrics, also not of the production waste (bag produced in Germany). Case relevant for all PU coated fabrics (of VAUDE)	Typical outdoor gear consumer is very environmentally minded. VAUDE wants recyclable products as a world first for its clients. VAUDE already has a strict policy on corporate social responsibility and also has its internal 'Green Shape Standard'[81]	Quantities of a typical product line are ca. 10.000 pieces/year, resulting in approximately 30t/year. DECOAT technology relevant for several product lines: bike bag, several backpacks, rain coats,… VAUDE manufactured in total over 2.5 million products in 2016, total turnover ca. € 100 mio. **DECOAT relevant on mid-term for ca 35% of turnover.**
PANASONIC - Switches & sockets 	PC, ABS and PC-ABS as substrate materials. Three types of 'coatings': (i) Foil Print with alu foil (ca 50µm); (ii) Chrome coating via dipping (primer of copper, chrome top layer); (iii) Painted layer, based on styrene-acrylic emulsion (ca 15µm)	PANASONIC wants to produce a 'green line' of products that are fully environ-mentally friendly without compromising on the aesthetics.	PANASONIC produces ca 60 tons of coated plastic switches, sockets and other coated electronics parts. **DECOAT approach expected to be integrated in ca 80%** of them within 5 years after project end

81 http://csr-report.vaude.com/gri-en/product/greenshape-concept.php

MBT - Cockpit part 	Substrate material is a polyurethane (PUR) foam. Coating is an ABS foil. ABS foil mechanical recycling and PUR chemical recycling	Both parts are currently not recyclable. Up to now, no regulation for ELV in case of buses and trucks. MBT wants to be pro-active and engage in increased recycling	MBT manufactures yearly ca 4000 busses. Plastic parts make up ca 15% of the weight. Expectation is to have **at least 50% of coated parts using DECOAT solutions**.
MAIER – exterior automotive part: Rear door garnish	Material: ABS Minimum-Maximum process temperature: 210-250ºC Size: 900mmx200mmx50mm Total weight: 400 gr (2% coating) Coating thickness: ca. 50 µm vs thickness of the plastic 2 mm Coating: PUR (acrylate), primer+Colour+Clearcoat	MAIER wants to validate a coating for automotive sector that can be easily debonded from the plastic substrate. This will make decorated components easier to recycle so the materials can be reused for the very same applications, which is currently impossible because even the smallest amount of impurities produce surface defects on aesthetical parts.	For each part, expected sales are typically ca. 200.000 pieces. Expectation: 3-4 parts can integrate DECOAT technology within 5 years after the project, corresponding turnover in the range of € 5mio. Turnover of MAIER is expected to reach €385mio in 2018, a strong growth up from €296mio in 2016. On longer term, DECOAT results are relevant for up to 20% of MAIER turnover, corresponding to roughly €90-100mio, taking into account the expected continued growth.
MAIER – interior automotive part: Fascia	Material: ABS/PC Min. and max. process temperature: 210-250ºC Size: 230x280x80 mm Total weight: 200 gr (1,5% coating) Coating thickness: ca 30 µm vs thickness of plastic 2 mm Coating materials: thermoset PU (acrylates). One layer. Colour.		

The market creating potential of DECOAT stems from the systemic innovation of making coated textile and plastic parts recyclable. This means novel markets for triggerable additives (for partners RESCOLL and DEVAN), for optical monitoring systems (IRIS) and for the recycling systems (CTECH). This implies that on medium to long term a new novel value chain will be created for recycling of coated parts, building on initial DECOAT developments.

For realising the DECOAT impact, a crucial part is that waste streams with triggerable coatings have to be made available in suitable form and in sufficient amounts. There are two main inputs: production waste and EoL post-consumer waste. We have 3 streams to consider: coated textiles, consumer electronics components and automotive (see table with overview).

Waste stream	Production waste	EoL waste
Coated textiles	(roles of) fabrics , scraps → Directly processable	Textile-based products, → Need removal of accessories (buttons, zippers,…) Crucial: collection logistics
Consumer electronics	individual parts / half-finished products at OEM →Directly processable / some disassembly	Full products → Need shredding/ disassembly Already commonly collected
Automotive parts	individual parts at Tier 1 →Directly processable	Car parts & ASR → Need disassembly Crucial: improve logistics for high(er) degree of disassembly of target valuable parts

For production waste, there should be no issues: triggerable coated items should be collected separately, which is no problem as manufacturing companies have now already several separate waste types they collect. **Logically, it will take some years before sufficient products with DECOAT-based technology reaches the EoL stage**. The real strength of the DECOAT technologies is exactly that it is also suited for EoL because the triggerable coated waste can be detected and thus processed separately.

We illustrate the case of coated outdoor textile products. On short term, **partner VAUDE will offer its clients the option to send back its EoL products** that include DECOAT technology. That way, items will be grouped together and can be recycled. On medium term, collectors of EoL textile (like group De Boer Recycling Group – ASG member) will have to install the optical detection equipment for triggerable coatings and thus separate this waste stream. Then, they can send it to recyclers (transitional phase) or recycle it themselves (longer term). This will require them to extend their current recycling equipment with equipment based on the DECOAT recycling plant.

Also **for the recycling of coated electronics parts, an extension of the current waste processing equipment to separate triggerable coatings and to channel that fraction to DECOAT recycling lines**. This will be implemented by players like REMONDIS (member of DECOAT ASG). Full recyclability of its products will be **a competitive advantage for PANASONIC**.

For the automotive parts, the situation is more complex and DECOAT has to rely on supporting developments in vehicle disassembly. Indeed, currently only a small fraction of the vehicles is dismantled, only hazardous components or very valuable ones. The rest is all together shredded and separated. Focus in that process is on the recycling of metals, the rest is called the ASR or automotive shredder residue (see also 1.4.2) and is a highly mixed stream, containing also (some) shredded coated scrap. To avoid low yield and need to separate shredded fractions of very mixed composition, our recycling technology should be fed with disassembled coated vehicle parts. Although the latter is not evident at the moment, there are good foresights that this will be the case in the future. **DECOAT will help partner MAIER to be prepared for this novel market situation.** Furthermore, DECOAT will seek synergies with the project MultiCycle, just approved in the SPIRE10 call, coordinated by IRIS and counting several partners in common, where similar challenges will be faced in terms of identifying and dismantling specific (in that case composite) parts prior to recycling. For support in this matter, we have included car dismantling company GBP in the ASG.

Furthermore, an ongoing development is the robotic disassembly of cars driven by the need for a more thorough car disassembly, especially with the growing share of electrical vehicles[82]. In fact, the fraction of metal is decreasing in cars, so there is a need to recycle more of the other fractions to keep with the increasingly stringent ELV directives. The latter is often only feasible if the (valuable) parts are dismantled beforehand. Manual dismantling is not evident, therefore efforts are ongoing for robotisation, focus on removing electronics parts. Such logical development in sustainable manufacturing and production will support the acceptance of new processes that could be needed for DECOAT products' EoL.

A special case is formed by HDOR (Heavy Duty and Off Road vehicles), which includes busses and trucks. There, disassembly and remanufacture is already more installed than for standard (light) vehicles. For example, more than 93% of all the materials in a standard DAF truck can now be reused and **the plastic parts of a DAF truck can easily be separated during dismantling[76]**. **DECOAT results will help partner MBT who is specialized in bus manufacturing, and via the mother company DAIMLER also in trucks, in working further in this direction.**

Europe has a leading position in high quality, engineered items like coated textiles and plastics. Next to the actual DECOAT partners, we want to make the wider European textiles, plastics and coating and adhesives industry aware of the novel possibilities offered by the triggerable coatings and debonding and helping them to make the transition. For easing this transition, DECOAT will develop the eco-design tool, which means that the service providing SME partners IRES and ECOMATTERS will get a chance to boost their activities. Consultancy will also be given by the DECOAT RTD partners (CTB - textiles, AIMPLAS – plastics & recycling sector, IVV, CTAG- automotive and NTUA – microcaps, nanomaterials). For example, AIMPLAS will be well-placed to offer engineering and consultancy to recycling companies wanting to adopt DECOAT technologies, via upgrading their current recycling facilities, or via building new ones. We will prepare the EU industry to switch to the next generation of coatings that can debond on demand and help them to implement this all along the value chain. We will **give the European textiles, plastics, coatings and recycling sectors an important and lasting competitive advantage via triggered debonding solutions**.

We will specifically engage in close collaboration with other related projects (such as MULTICYCLE, see also above and Task 7.5) and look into common publications/events to **raise awareness towards relevant target groups on the emerging recycling possibilities for coated textiles and plastics.**

2.1.3 Environmental and social impact – policy framework

Environmental impact of DECOAT is tapping into the sustainable development policies of the EU, which is committed to a 'high level of protection and improvement of the quality of the environment' (Article 3 of the Treaty on European Union). As it has been shown in section 2.1.1.4, the suggestions of the DECOAT project can facilitate reduction of greenhouse gases emissions and amount of the generated waste to be landfilled or incinerated. The potential for waste generation reduction is **ca. 2.1 million tons for automotive parts, ca 200 kilotons for electronic goods, and ca 175 kilotons for coated textiles.** That refers to including

82http://www.centreforsmart.co.uk/system/publications/attachments/000/000/183/original/Robotic_disassmbly_for_increased_recovery_of_strategically_important_materials_from_electrical_vehicles_04102017.pdf (2017).

these amounts of the recycled material to the material recovery level of the waste management pyramid compared to the lower levels of energy recovery and landfilling. This is especially relevant in view of the current EU waste policies and regulations (see 2.1.4, including an overall trend for developing circularity and for landfilling ban in some European countries (such as presumed by Austrian and German Landfilling Ordinances). Moreover, the estimated amounts of material recovery have a very large GHG emissions reduction potential. Under Article 191 of the Treaty on the Functioning of the European Union, combating climate change is an explicit objective of EU environmental policy. Under this goal, EU is pursuing the targets of reducing GHG emissions by 40% by 2030 and by 80% by 2050[83]. This significant contribution to reducing global warming potential of manufacturing activities in Europe could be also complimented by the reduction of the GHG emissions coming from the incineration or landfilling of the coated parts and materials. Such for example, the avoided emissions from coated plastics recycling (assuming PVC reduction potential for all coated plastics) could be estimated equal to 175*1.27=222tons CO2eq.

A recent study conducted by Material Economics[84], shows how circular business models, re-circulation of materials and greater material efficiency of products are the keys to build a competitive industrial economy with zero net emissions in Europe. All these keys are being addressed by DECOAT. To achieve this objective, **three types of circular strategies** associated to each other have been envisaged to reduce EU industrial emissions: (i) the reuse of different materials, (ii) the reduction of the amount of materials that are discarded in the manufacturing or construction process, (iii) the creation of new business models that reinvent the way we provide services and goods. In practice, these broad strategies translate into a wide range of measures: changes in product design and materials choice, improved technology, modernized value chains, etc. The **benefits of circular approaches are strengthened by the fact that many of these measures are complementary and that many of these abatement opportunities are economically attractive**. A more circular economy would have many other benefits as well, such as reduced geopolitical risks, local job creation, lower air pollution, and reduced water use.

Social impact of DECOAT is based on the reduction of waste generation and of the system carbon emissions into human health and general well-being improvements. Basic regulations regarding H&S issues include arrangements for implementing the health and safety measures, identified by the risk assessment, set up emergency procedures, provide clear information and training to employees. Moreover, recycling initiatives and businesses based on technical solutions and business models suggested by DECOAT will increase GDPs and create work places (respective to the EU policies in employment and labour market) and stimulating local economies (Small business Act). The suggested DECOAT recycling schemes have a high potential to tap into the EU Social Economy framework (Social Business Initiative, Social Innovation policy making, Single Markets Acts 1 and 2). Additionally, the GHG emissions reduction due to the implementation of DECOAT solutions has a potential to avoid otherwise necessary measures for climate change adaptation, including in local communities and groups with high vulnerability.

DECOAT results will help to achieve several European policies, in waste reduction, climate change action and mitigation, and social impact (see also section 2.1.1). **DECOAT also fits in the Circular Economy Package** consisting of an EU Action Plan for the Circular Economy that establishes a concrete and ambitious programme of action, with measures covering the whole cycle: from production and consumption to waste management and the market for secondary raw materials. The **DECOAT developments on automated sorting fit the EU strategy for digitisation the manufacturing industry.**

2.1.4 External conditions influencing the expected impact

Current low market demand for recycled plastics, which is limited to 6% of plastics demand in Europe is discouraging private investments in recycling capacities. In the EU, uptake of recycled plastics in new products is low and often remains limited to low-value or niche applications. One of the reasons for the low use of recycled plastics is the misgivings of many product brands and manufacturers, who fear that recycled plastics will not meet their needs for a reliable, high-volume supply of materials with constant quality specifications. The **European Commission will promote the development of international standards to boost industry confidence in the quality of recyclable or recycled plastics**. It will also explore more **targeted sectoral interventions in key sectors with high potential like packaging, construction and automotive**.

Legislation and regulations on recycling, re-use and waste form important drivers, potentially further enhanced by public procurement. There are several European directives on these issues that directly affect the production methods and materials choice of electronics consumer goods and cars. For the DECOAT sectors, following directives are crucial:

- *Waste Framework Directive* (Directive 2008/98/EC) aims for a European recycling society with a high level of resource efficiency by increasing re-use and recycling of waste from households to a minimum of overall 50 % by weight by 2020.
- *Waste Directive – Electrical and Electronic Equipment* (Directive 2012/19/EU). Waste of electrical and electronic equipment (WEEE); eg computers, TV-sets and fridges is a fast growing waste streams in the EU, expected to grow to more than 12 million tonnes by 2020. Overall aim is to recycle at least 85% of electrical and electronics waste equipment by 2016 in EU.

83 https://ec.europa.eu/clima/policies/strategies/2050_en
84 https://www.ellenmacarthurfoundation.org/circular-economy

- European Directive 2005/64/EC on the type-approval of motor vehicles with regard to their reusability, recyclability and recoverability) state that 85 % of the mass of the vehicles included in the M1 and N1 categories (passenger cars and pick-up trucks) must be recyclable/reusable.
- With respect to heavy duty vehicles, including busses and trucks, there are currently *no* End-of-life regulations that impose mandatory recycling and/or recovery targets[76].

These directives (or the lack of) pose challenges (and opportunities) for the automotive and consumer goods sector. **The DECOAT developments will help PANASONIC, MBT and MAIER to use renewable materials in (parts of) their products, thus working towards the directives.**

The Circular Economy Action Plan[85] aims at improving the uptake of secondary raw materials. **DECOAT exactly aims for this by overcoming existing barriers for coated plastics material**. DECOAT goal is to enable recycling and thus re-use of raw materials used for making coated parts. Thus, DECOAT is in line with the **7th Environment Action Programme[86] and with the recently published 'Roadmap to Resource efficient Europe[87]'** which aims at promoting waste recycling and re-use.

In the context of ongoing and upcoming evaluations of EU rules on construction products and on end-of-life vehicles, the EC will look into specific ways of promoting this. In the context of future work on the Packaging and Packaging Waste Directive, thought will also be given to using economic instruments to reward the use of recycled content in the packaging sector. Finally, the Commission will work on integrating recycled content in Green Public Procurement criteria (European Commission, 2018).

Generally, these directives and other (national, regional) legislation can have a strong positive influence on market opportunities for use of recycling content. As an example that will be applied to other sectors: As a result of the implementation of the Plastic Bags Directive (Directive (EU) 2015/720) some EU Member States have already started to set requirements for the uptake of recycled content in plastic carrier bags. For instance, the Spanish Royal Decree 293/2018 establishes that from 1 January 2020 plastic carrier bags with a foil thickness larger than or equal to 50 µm shall be made from at least 50 % recycled plastic.

This shows that **DECOAT will place the participating companies in pole position for when the (legislation enforced) need for more use of recycled material becomes even higher**.

Evolving recycling landscape

However large the need and however impressive the targets, **today only 6% of new plastic materials used stems from recycling**[88]. Also within textile industry the use of recycled material is limited: for clothing only about 1% is closed loop, ca 12% cascade recycling[3]. A great deal of work is underway to improve this (also via European projects), focussing on technical, economic and logistics aspects. As a result, **we assume in the near future well-established networks throughout Europe where waste is collected and brought to local recycling facilities**.

Current recycling schemes for plastics from WEEE and ELV treatment rely mainly on post shredder treatment approaches since only large scale products are subjected to manual dismantling due to economic reasons. Coated products of concern in the DECOAT project might be too small (e.g. switches) or too inaccessible (e.g. dashboards) for manual dismantling. Thus, identification of coated plastics in plastic shred by automated detection systems will be key for future access to coated plastics. **DECOAT will therefore investigate optical identification tools for coated items to be implemented in sophisticated WEEE or ELV sorting technologies (WP3). This technology will ensure the future access to EoL products containing DECOAT ready-to-recycling coating systems.**

Textiles and especially complex textiles as backpacks are not recovered in specific waste streams and currently end up in landfills (if allowed) or incineration plants. However, the high durability of such products qualifies them for an EoL treatment by re-use or repair companies. Such entities are intended to be future sources of complex EoL textile waste such as the ones addressed in the DECOAT framework and which will be investigated in detail.

Reluctance of adopting recycled materials in the industry because of specific processing needs, lower performance and/or early negative experiences. Several companies did already first trials with recycled materials and experienced difficulties in their processing, e.g. on reproducible quality levels or in processing. These first negative experiences, even if not valid anymore with the current materials, often remain and keep people biased towards further trials with recycled materials. The DECOAT technology providers are aware of this perception and counteracting on it will be a key part of the message brought during dissemination. One way to promote the DECOAT technology and the recycled material will be via the public eco-design tool, see Task 7.5.

Cost of novel solutions and cost for recycling process/recycled materials

DECOAT solutions will mean that the coatings will be more expensive at initial production phase (ca 10-15% as rough estimate). This is often rather small fraction of the total cost of the product, eg the bike bags. Further, gain will come because of less waste,

85 European Commission: Circular economy Action Plan – closing the loop
86 European Union: Seventh Environment Action Programme
87 European Commission: Roadmap to a Resource Efficient Europe.
88 https://ec.europa.eu/commission/sites/beta-political/files/plastics-factsheet-industry_en.pdf

ability to reuse materials. Therefore, rely on the emerging new business models and schemes like EPR – Extended Producer Responsibility, increase cost for waste disposal, etc.

General economic situation. The automotive and household appliances sectors are cyclic sectors, implying that they strongly depend on the general economic situation. This can influence the sales foresights, especially for MAIER, MBT and PANASONIC. Currently, economic conditions in the US and Europe are high but in the future a higher market fragmentation for the industry due to the emerging economies is coming into play, making innovations even more important. Companies driven by innovation in the area of circular/recycling will beyond doubt benefit and maintain market share. Therefore **the coated parts recycling solutions offered by DECOAT will help to become more resilient against economic fluctuations**.

2.2 Measures to maximise impact

(a) Dissemination and exploitation of results

To ensure successful exploitation of the DECOAT results, **dissemination and exploitation activities will run in parallel with the project implementation**. Integrity and consistency of all dissemination and exploitation efforts will be supervised by the Exploitation and Innovation Manager (OSM, see also section 3.2.1.1b). The strategy for these activities is as follows:

- **Dissemination**: all partners will collaborate in dissemination of results among the relevant industry sectors;
- **Exploitation**: industrial take-up of the materials and related processes developed in DECOAT plays a central part: each partner will be involved in defining a preliminary business plan for its key exploitable results (KER).

WP7 includes tasks dedicated to the dissemination and exploitation of the project achievements. The Plan for the Exploitation and Dissemination of Results (PEDR) will be completed at the start of the project (preliminary version of D7.2, see WP7), improved and evaluated through the duration of the project. It will **ensure that all project partners have a joint understanding of dissemination and exploitation and which knowledge protection measures will have to be taken**.

The draft PEDR

The draft PEDR presented in this section will be further elaborated from the start of DECOAT and regularly assessed and adapted to the actual evolution of the project, see the related (incremental versions of) deliverables that will be delivered in WP7. We will also follow and implement as much as possible the guidelines as recently published by the EC on social media: '**Social media guide for EU funded R&I projects**'[89].

A. Target audience and message for a complex network of stakeholders.

Disseminating project results at working groups and strategic committees beyond the consortium will take place. We will address the following groups of stakeholders to maximise the impact: plastic/textiles/coatings converters, end users, waste managers, recyclers (to generate a push/pull effect across the value chain) but also policy makers and the wide public (to promote the EoL products' collection and promote innovative public procurement for DECOAT solutions).

B. Channels for dissemination: from public online to interpersonal face-to-face interaction.

Dissemination will be based on three major channels: (i) **Online dissemination** – website and social media, webinars, electronic newsletters, and digital press releases; (ii) **Offline dissemination -** classical vehicles of knowledge transfer like project workshops, publications, conferences and trade fairs, printed factsheets and flyers; (iii) **personal direct dissemination** via the formal and informal (personal) networks of DECOAT partners. **A list of key dissemination channels specific for each partner has been collected** (see the partner descriptions in section 4.1).

C. Preliminary overview of dissemination actions: continuous actions and dedicated campaigns.

To disseminate as efficiently as possible (in terms of reach and cost), two main pillars are defined: (i) continuous dissemination relying on existing events and means and (ii) organising dedicated DECOAT dissemination campaigns and events.

(i) Continuous dissemination.

Goal: Raise awareness and interest, ensure that 'external stakeholders' remain aware of DECOAT, keep them committed and ensure they can follow the project progress. Creating opportunities for direct interaction with interested stakeholders, regular newsletter for sharing project news and announcing conferences/events. Continuous activities also include: updating project information on the DECOAT website and on social media platforms, distributing leaflets,… Thereby using the materials and channels established in DC1 (see below).

Audience: with this regular stream we want to reach as wide a public as possible, including key stakeholders and also the wider public, depending on the content.

Key message: broadcast of smaller news items, updates, events, … but also more extended updates on the progress of the value chain demonstration that will be implemented in DECOAT.

89 http://ec.europa.eu/research/participants/data/ref/h2020/other/grants_manual/amga/soc-med-guide_en.pdf

Organisation: ISWA (NL) as WP7 leader will coordinate. Dissemination material will be collected from ALL PARTNERS and a schedule will be set-up to efficiently organise the dissemination, leveraging synergies and avoiding any possible overlaps.

(ii) Dedicated DECOAT Dissemination Campaigns (DC)

Goal: the Dissemination Campaigns (DC) will give a boost to DECOAT's dissemination efforts at crucial times in the project schedule. The goal is not to depend on external events to be able to promote the project but to proactively choose our own stages and adequate timing. This requires the ability to garner the attention of and create impact with a significant number of relevant stakeholders, and the consortium is confident to be able to create this momentum for its campaigns given the ASG and the associations within or closely associated with the consortium and the extensive existing dissemination channels. Because the dedicated Dissemination Campaigns (DC) form the key element of the activities, they are described separately below.

DC1 – create awareness. *Goal*: to create awareness for and stimulate interest in DECOAT based on the ambitious goals, innovative nature, commercial and sustainability potential, and the diversity and reputation of the consortium members.
Target audience: ASG, materials researchers, potential end-users, the coating community, the general public, policy makers, in short: very wide and covering all actual and potential target audiences.
Activities: the dissemination activities will start immediately through a press release linked to the kick off meeting. Further key activities are performed through the setting up of the DECOAT project website and having first dissemination material available (e.g. DECOAT leaflet -digital and print), all following a distinct project visual identity. Furthermore, the consortium will make extensive use of consortium partners' dedicated professional social media channels, especially LinkedIn and Twitter.
When: by Month 5 - *Responsible partner*: ISWA (NL) and CTB (as coordinator)

DC2 – call for collaboration. *Goal*: Call for collaboration to continue and expand on DECOAT results that need external partners or licensee to be exploited, i.e. results stemming from WP1-WP4 that can be applied outside the use cases of WP5.
Target audience: ASG, innovators/entrepreneurs, large enterprises with own R&D, innovative SMEs, related current and future projects, coating/plastic converters, recyclers, waste management bodies, product designers and developers.
Activities: public events organized by DECOAT, compilation of a white paper on DECOAT coating debonding solutions.
When: max 4 months after MS5 (M31 - end of main development work in WP1-4) is reached (M35) - *Responsible partner*: CTB

DC3 – promotion of DECOAT results. *Goal*: Promotion of project results (focus on new solutions for recycling of coated parts) to pave the way for their further development, exploitation and commercialisation.
Target audience: materials and coating community, current and future research consortia, innovators/entrepreneurs, plastics converters, textile industry, end users, policy makers, standardisation bodies and regulators, general public.
Activities: press releases, publication of peer-reviewed and other articles with the goal of promoting project results with a clear focus on the use cases. Dissemination via selected conferences and own DECOAT events. Storytelling as format for presenting the prototype cases, highlighting the promising role of the novel DECOAT solutions. By specifically targeting end users, investors and policy makers, DC3 will pave the way for their adoption, commercialisation, and further development.
When: by M46 - *Responsible partner:* ISWA (NL)

An overview of dissemination actions in given in the table further below (pag 32). In section 4, for each partner its specific dissemination channels, like conferences, networks,… are listed.

D. Exploitation activities: identification of results, outline of potential and steps to bring to market

D1 Identification of key results

The **table below (pag. 34) shows the overview of the Key Exploitable Results integrated along our partner's value chain and initial exploitation also in link with previously discussed dissemination steps**. The results most suited for direct industrial exploitation are further elaborated in part D3 – Steps to market and preliminary business plans below.

D2 Outlining of the potential market

Plastic Recycling and recycled plastic market: The global plastic recycling market has been gaining a steady momentum over the past few years due to the related directives and growing awareness about CO_2 emissions and the need to reduce them. This, together with the overflowing state of landfills has also triggered an urgent need for recycling plastic instead of producing virgin material. The global plastic recycling market, valued at **€25 bn in 2015** (of which only €2 Bn in EU[90]), is expected to reach a figure of **€45 bn by 2024**. During the forecast period of 2016 and 2024, the global market is **expected to grow at a CAGR of 6.9%**. Steady growth is expected in the global market due to ever more stringent laws and increasing environmental awareness, especially in EU. In 2012, out of the 25.2Mt of plastics collected for recycling, 6.6 Mt were recycled in EU and 8.9 Mt were incinerated. Incineration is massively used, with for example 38.8% of plastic packaging waste[91]. Currently, according to the

90 Coltelli M.B., Aglietto M., Riutilizzo dei materiali polimeri, Edizioni Nuova Cultura, 2015
91 Plastics Europe, association of plastics manufacturers : Plastics, the facts 2017

Plastic Recyclers Europe, 1000 companies are operating with 30.000 employees and a 3Mt installed capacity. Currently, the **European installed plastic recycling capacity is practically being fully used and mostly focused on packaging, indicating the huge potential for new investments for and adopting new (more) efficient technologies, provided they are proved economical viable**.

We calculate that DECOAT solutions address ca 3.5 million tons of coated waste material (see section 2.1.1.3). For calculating our direct impact, we assume that **we reach on longer term 8% of the potential market**, i.e. ca 280ktons. This means that **DECOAT can improve the competitiveness and innovation capacity for a market segment of ca €1.7billion** (assuming on average a commercial value of ca €6000/ton for these coated materials). **For reaching a market share of 8% we count on the market pulling effect of the 'brand owners' in DECOAT** (Daimler group via MBT, PANASONIC and VAUDE) and **on the ASG members**, especially on the leveraging effect of the industrial associations and networks. These will help to motivate companies outside DECOAT to adopt our technology. **A natural entry to the market will be addressing production waste**. Nowadays at industrial level **ca. 10-12% of the freshly produced coated parts are considered production waste** because of the combination of complex coating/painting processes and very high-quality standards, especially in automotive, electric electronic and transport industry. These parts cannot be reused because they are 'contaminated' with the coating material. They can be sold at less than €100/ton for very low added value applications or, even worse, be discarded of for land-fill or thermal recovery. This is an important loss factor because the commercial value of the parts is estimated at ca. €6000/ton. Estimating the actual material value is difficult as the coated parts concerns both expensive engineered polymers as well as less valuable materials. On average, we estimate their value here at €2000/ton. Given the easy material availability (see also above) and an estimated recycling cost of ca. €500/ton (for logistics, triggered removal and recycling using CO2 assisted extrusion, see also section 1.1.1.1)**, DECOAT can offer a viable solution for recuperation of coated production waste delivering cost effective secondary materials.**

DECOAT will create a market need for the triggerable additives. Starting again from the 280ktons of coated waste material, and assuming that on average a coating or primer layer represents 3% of the total part weight and that we need to add 5% of triggerable additives, we calculate a need for 420tons of additives per year in Europe. At an average cost of ca. €50/kg, this means a **DECOAT triggerable additives market of over €20 million annually, or ca. 60 jobs**.

Implementing the **DECOAT technology will significantly boost the recycling sector**: if we consider again the recycling cost of €500/ton, this corresponds to **ca €140million extra annual turnover, or ca 400 jobs, created by enabling the recycling of these (280kton of) coated parts**.

To realise this, **recycling lines will need to integrate (or be updated in case of existing ones) with the optical detection technology (estimated at ca €150k/unit by IRIS), the MW technology (estimated at €70k/unit by CTECH) and/or the Creasolv technology (estimated at ca. €2000k/unit by IVV)**. Assuming a typical recycling line has a capacity of 5kton/year, approximately 60 lines are required to process the target material of 280kton. All lines will need the optical detection system, leading to a market of €9million (60 x 150k€). We assume that 30% will use 'standard ovens' (no direct DECOAT benefit), but 50% of the lines will install MW or other heating equipment, so a market of €2.1million (30 x €70k) and roughly 20% a Creasolv system, leading to €24million (12 x €2000k). All this sums up to **a one-time market of ca. €35million**, **which will create a recurrent annual turnover of ca €7million (20%) thanks to related services (consulting, maintenance, updating).**

SWOT analysis for DECOAT. Relative competitive strengths and weaknesses for the DECOAT applications have been analysed according to a preliminary SWOT Analysis that will be further elaborated and finalised within DECOAT.

Strengths	Weaknesses
Novel set of triggerable techniques for debonding Unique mix of interdisciplinary skills and expertise Strong industrial participation, including R&D driven SMEs	Cost of coating will increase → need to offset this by focussing on total cost (and potentially into new business models) for commercialisation
Opportunities	**Threats**
Answer the unmet demand for recycling of coated parts Grab potential of techniques that have proven merits for other applications, e.g. microwave additives Contribution to EC regulations concerning Plastics Strategy DECOAT solutions are transferable to other applications, e.g. in construction, composites, furniture, also for enabling repair	A systemic approach is needed, might be difficult to introduce the novel concepts → counterbalanced by the strong pressure to recycle plastics, see e.g. the EC initiatives Low virgin material prices, depending on oil price → on longer term continuing low oil prices are highly unlikely

Table: overview of planned dissemination activities and link with key dissemination elements: dissemination campaigns "DCN" (DC1, DC2 and DC3), continuous communication ("Cont.") and Advisory & Support Group ("ASG").

Activity	Goal	Concrete action	Lead / partners	Timing	Target group	Expected impact
Website	Regular updates, news section, partner info, event listing, etc.	Design and formulate contents, keep up to date	ISWA (NL), CTB	Operational by M4, cont.	All stakeholders	Central repository of project information; communication materials direct back to here to achieve continuous engagement.
Workshop and webinars to selected stakeholder groups	Minimum of 6 workshops to 3 types of audiences in 3 regions, with 15-30 participants. At least 4 webinars with 25 participants each.	Identification and acquisition of audiences, creation of workshop and webinar contents based on the needs of different target groups, delivery of workshops. Selection of most suitable speakers from among the consortium, market to relevant potential audiences, deliver to as many participants as possible; post-event distribution, make available on website	ISWA (NL), all but primarily RTDs & industrial	DC2 & 3	All stakeholders	Boost for exploitation as these are very specifically targeted to audiences known to take a strong interest in the project. Also feedback for project to improve its processes and dissemination. Cost-effective in-depth dissemination of project results to diverse stakeholder groups.
Final public event	One event, at least 75 participants	Organisation of professional high-level 1-day conference on the novel DECOAT debonding solutions and materials	ISWA (NL), NTUA, all partners	DC3	International high-level attendees	Multiple key contacts and avenues for post-project exploitation and uptake.
Dissemination of public deliverables	As appropriate, i.e. linked to workplan deliverables	Creation of reports or other documents or actions as required by the deliverables Led by the partner in charge of meeting the respective deliverable	ISWA (NL), all partners, as per the workplan	DC1, 2 & 3	Mainly research / project community, ASG	Engage research community and provide evidence of the potential and opportunity connected with DECOAT
Flyers, brochures, factsheets, white papers, etc. - print and digital	Public dissemination, tailored to specific groups such as scientific, policy makers, etc. 2,500 prints to be distributed.	Creation of visually attractive and professionally written documents.	ISWA (NL), CTB, all partners	Flyer by M5, others based on developments in DC 1 & 2	All stakeholders, ASG	Easy-to-grasp general or condensed specific information to raise awareness, stimulate interest, and get a general audience to engage with the project

Social media, e.g. YouTube, Twitter, LinkedIn, ResearchGate	Regular posts (monthly); targeting >5,000 followers	Regular posts via all available channels; engagement and interaction with followers and readers.	ISWA (NL), CTB, all partners	Cont.	All stakeholders	Initial awareness raising, broad but 'superficial' network, spreading news widely quickly, reinforcing all public dissemination activities.
Coverage in mainstream media	At least 3; this can be newspapers, radio shows, science-focussed TV shows.	Liaison with media outlets and submission of material to them, e.g. in the form of articles, interviews, videos scripts, in-person appearances	ISWA (NL), all partners	DC 3	General public	Create awareness among the general public and stimulate interest in the sustainability potential of recycling coated parts.
Articles in peer-reviewed journals and magazines	minimum of 5	Liaison with the most relevant media. Drafting and submission of professional, ready-for-submission, scientific articles.	RTDs, ISWA (NL)	DC2 & 3	Scientific audience	Disseminate project results to potential adopters for exploitation in future research.
Press releases	Every 6-12 months. Distributed across the media channels of all partners.	Write informative and well-edited document with links to relevant websites; distribute to all accessible media networks through the partners and via other communication channels, e.g. website	ISWA (NL), CTB, all partners	Cont., but especially DC2 & 3	All stakeholders	Key for dissemination through 'traditional' media outlets.
International trade fairs and conferences	At least 1 trade fair and 3 conferences per year, including presentations, exhibition, sponsorship /media partnerships	Keep track of all relevant events, prioritise among them and coordinate partners' attendance; arrange for exhibitions and/or sponsorships where feasible and relevant, and push for speaking contributions. Coordinate with other initiatives/projects	ISWA (NL), all partners	DC2 & 3	Scientific, business, industry stakeholders	Meeting and presenting to the most relevant exploitation target groups for a direct innovation and knowledge transfer
Outreach to local communities, schools, and universities	At least 3 information events or opportunities to share project information, targeting 100 students, citizens, or pupils.	Connect with relevant organisations such as schools, universities, consumer or citizen organisations, NGOs, public events; provide details about project, suggest talks and information that could be disseminated to respective audiences.	ISWA (NL), all partners	DC2 & 3	General public	Create awareness among the general public and stimulate interest in the sustainability and recycling of coatings.
Leverage existing channels: memberships, company newsletters, etc.	Several times/year	Assist partners with disseminating project information among their networks; potentially translation of materials into languages other than English; identify key networks into which to disseminate and organise input of project information with the respective consortium partner associated to the network.	ISWA (NL), all partners	Cont.	All stakeholders	Mobilising existing networks to create visibility for DECOAT, disseminate to organisations already working with our partners and likely to have matching interests, exploitation potential within partners' peer groups.

Table: overview of Key Exploitable Results.

Description of Exploitable Result	Main Partner(s)	Our specific improvement vs the current status	Exploitation: what is the (market) need / business opportunity it responds to ? Which strategy to follow ?	Potential users (outside consortium) and timing	Activities to reach users – link to DECOAT dissemination actions
Triggerable debonding system for removal of coatings from textile and plastic substrates	CTB, AIMPLAS, DEVAN	Currently there exist no practical technologies for removal of coating, only limitedly via chemical dissolving. Targeted release of coating via a trigger in an economic and environmental way	No current method to recuperate coated textiles For first time enable the recycling of coated textiles Offer technology via licensing	(Large) companies specialised in coating and coating formulation producers (for substrates outside scope of DEVAN) Within 1st year after DECOAT ending	DC2 – find partners for licensing Present at coating fairs and conferences
Recycling pilot line for the triggered removal of coatings	AIMPLAS, CTECH, IVV	Selection of components to increase the efficiency, speed and quality of the plastic separation. Knowledge on plastic recycling & pilot plant elements optimization.	Use of MW technology for recycling of coated parts Generate corresponding technical assistance and technological transfer to interested parties. Commercial agreements. Intellectual property. Technology transfer	Recycling companies wanting to extend their capabilities to also recycle coated parts	During DC3 event, a visit to the recycling line will be planned in
Materials needed for the triggered debonding:					
•Thermo-responsive core-shell nanostructure d co-polymers	NTUA, DEVAN	A variety of core-shell polymer nanoparticles can be found in literature. Use in waterborne formulations for textile industry. Function under specific stimulus (temperature).	Novel debonding technologies for the removal of coatings over plastics and textiles will be developed with the use of nanostructured co-polymers, functioning under thermal triggers. Core-shell co-polymers will be combined with dyes to enable "smart sorting" in the recycling industry. Expected sales price: €40-50/kg Possible Patent is foreseen. Licensing or, if strong enough IP, **potential for start up creation**	Coating (ingredient) producers Extended use: Use as drug delivery systems (medicine sector); Use also in environmental clean up Market ready ca 5 years after DECOAT.	Scientific conferences and publications Presentations at coating and paint conferences Interact with via ASG member 'European Coatings' Journal
•Super Absorbent Polymers	NTUA	A variety of core-shell polymer nanoparticles can be found in literature. Commercial available, but no structured. Size of micrometers. • Use in waterborne formulations (steam and	Novel debonding technologies for the removal of coatings over plastics and textiles will be developed with the use of SAPs, which will operate under heat and steam triggers, as well as in selective solvents. Recycling industry will benefit since new processes with smart separation techniques will be implemented.	Coating (ingredient) producers Extended use: SAPs can be also used in other sectors such as agriculture, personal intensive care costumer goods, waste water treatment, medicine for drug-delivery systems, sealants.	

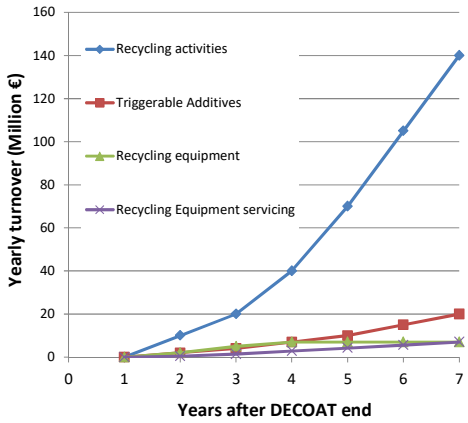
		selective solvents). • Nanostructured Low polydispersity index	Possible Patent is foreseen. Licensing.	Market ready within 2-3 years after DECOAT	
•MW& Rf assisted hybrid carbon-based nanomaterials	NTUA, CTECH	Heating in the research of carbon-based materials has been mainly limited to liquid solutions. Microwave heating of solid state carbon-based materials and especially CNTs, has distinct merits, but they have scarcely been reported (T. Kim, et al, 2014). Hybrid CNTs will be synthesised either grown on SiO_2 nanoparticles or they will be decorated with metal oxide nanoparticles to enhance the inductive heat due to MW irradiation.	CNTs are very strong mechanically and thermally, recycling should not affect their properties. Hybrid CNTs will have a competitive advantage in comparison to current commercial ones due to their multifunctionality. Possible selling price €50/kg (as hybrid materials). Establishment of **spin-off company** for the commercial exploitation of the hybrid materials. MW equipment: licensing to equipment manufacturers, equipment leasing to recycling companies. Expected cost per system depending on capacity – estimated at average of €70k/system.	Targeted users: Hybrid CNTs: the recycling industry, also composites and coatings sector. MW equipment: recyclers Hybrid CNTs market ready within 2-3 years after DECOAT MW equipment market ready by end of the project	Hybrid CNT: scientific conferences and publications MW equipment: fairs and industry events Present to networks concerned with waste and recycling, like WRAP in UK.
Debondable adhesives and primers	RESCOLL	Limited commercial products. No commercial primer available for debonding. Development of debonding primers and adhesives with processing and debonding capabilities (thermal triggering) adapted to the use cases.	New bonding solutions for recycling, maintenance and upgrade of bonded products. Improvement of disassembly and sorting of multimaterial assemblies. Sales of the adhesives and primers formulations	Automotive sector (OEM and TIER1), consumer goods, also aerospace and composite sector Ca 5 years after DECOAT as the durability of the debondable system must be proven before product launch	Attendance to fairs on bonding / coatings. Webinars on debonding. Commercial brochures and online information (webpage/blog articles). Production for new users (prospection work)
Monitoring systems **for optimal DECOAT materials processing and end of life sorting and recycling**	IRIS	Generic sorting systems using a number of techniques exist but they are not adjusted to DECOAT materials which are new. Looking at the most versatile/performant/cost effective combination of techniques, we will design and deliver optical systems that allow optimising DECOAT handling processes throughout its life cycle	Ability to actually identify and recycle the new material. Availability of secondary raw materials. Green image of the value chain. Quality control for the debonding / coating recycling. Sales of monitoring and sorting systems. Cost: will depend from case to case. Rough estimate of on average 150k€ (including the chemometrics model)	Recyclers of waste streams When the materials and recycling lines are becoming available on the market, IRIS would need in the range of 1 year to provide a system for a larger plant	Demonstration events at the pilot plant of AIMPLAS or IVV will help as it will be a practical showcase of the systems to engage potential new end users, eg during DC3 event

Novel application of the CreaSolv® Process	IVV	Dissolution and recovery of thermoplastics from composites and plastic mixtures. Application to coated textiles and special equipment in ELV and WEEE	Process might enable recovery of pure fibre polymers from post-consumer textile waste; Process might increase the share of target polymers of WEEE recyclers licence the process	Chemical engineering companies will build equipment, recyclers will run novel treatment plants Market uptake after further upscale within 3-4 years	
Recyclable coated parts: textiles for outdoor use; consumer electronics; structural automotive parts	VAUDE, PANASONIC, MBT	Recycling of coated textiles/plastic parts is so far not possible/ viable. First time a workable, cost competitive solution.	Circularity on coated textiles within our outdoor textile industry. Promote the novel DECOAT solutions throughout our supplier chain and stimulate/force them to adopt the novel technologies (market pull strategy).	End user clients – target group of highly environmentally concerned people, eg LOHAS[92] End-user suppliers: textile and plastic finishing and coating companies, worldwide Follow up project to bring to sufficient high TRL - within 2-3 yrs after DECOAT	Use of press releases at DC1 and DC3 to communicate to general public and the
Recyclable automotive aesthetic components	MAIER	A high percent of the aesthetic components of the vehicles are coated. There is no cost effective debonding process that could bring good quality recycled material for manufacturing aesthetic components, because the smallest impurities produce surface defects on aesthetic parts.	MAIER will validate a coating that can be easily debonded from the substrate. This will make decorated components easier to recycle so the materials can be reused for the very same applications, which is currently impossible. Carmakers strive to develop circular economy model. MAIER will increase its market share by offering aesthetical finishes that can be recycled. Sales of the coated parts.	Most of the carmakers established in Europe are clients, so potentially every automotive OEM is a customer. Sales: As soon as the DECOAT technology is ready to be industrialized (will require high TRL project – eg type FTI).	Participation in fair, promotion of DECOAT results during visit to clients
Ecodesign tool **for** opportunities identification and benefits assessment of recyclable coatings **(based on LCA, LCC and safety aspects)**	ECOMAT, IRES	Currently the existing eco-design tools do not cover the applications to be considered within the project. The DECOAT tool will allow to map the main impact hot spots of coating materials applied to a 'product' (i.e. coating & substrate) and will be able to identify the opportunities and benefits of implementing the DECOAT debonding technology.	The favourable results of the full assessments could be used by manufacturers to proceed with green marketing and communication strategies, as well as relevant certification and labelling of their products, leading to increased market shares due to better environmental performance. Release (basic version of) the tool in public domain, free to use by all but offer (paid) technical consultancy to use the (extended) tool optimally	Major manufacturers in automotive plastics, coated fabrics and coated electronics sectors. Within 3-5 years after DECOAT Extend to domains outside DECOAT Within >5yrs after DECOAT Dissemination of the tool use among the manufacturers has a high potential to facilitate the use of the DECOAT solutions in their practices.	The open version of the tool will be available for free use on DECOAT website. Communication of the tool will be implemented via main channels of the DECOAT partners, including newsletters, publications, fairs and conferences.

92 LOHAS = Lifestyles of Health and Sustainability, see e.g. https://en.wikipedia.org/wiki/LOHAS

D3 Steps to market and preliminary business plans

The DECOAT industrial partners, whose foreground should ultimately lead to industrialisation/commercialisation, have carried out preliminary business plan canvassing from which a compound brief is outlined below. This is based on the market creating potential (section 2.1.1.2), the available amounts of coated waste (section 2.1.1.3) and the outline of the market potential (part D2 of the PEDR) and taking into account the improved innovation capacity (see section 2.1.2).

<table>
<tr><th colspan="3">DECOAT BUSINESS CANVAS</th></tr>
<tr>
<td>Vision
Provide sustainable solution for introducing coated parts into circular economy.</td>
<td>Competitive advantage
Novel smart polymer material systems for debonding of coatings, applicable to the main industries - the respective recycling method and the related processing.</td>
<td>Target market
Coated materials industries that are impossible to recycle up to now</td>
</tr>
<tr>
<td rowspan="2">Problem: (multi-layered) coated parts not recyclable because of permanent bonding
Solution: Create triggered debonding systems for coated parts, the linked recycling process and the corresponding circular economy roadmap.</td>
<td rowspan="2">Marketing activities
Workshops with Industry Engineers and Technicians in order to adopt DECOAT materials and processes
Participation in exhibitions and events with booth and presentations
Market pull by ‘big brands’ & technology push by innovative R&D (SME) companies
Mass marketing to the general public to act as leverage for industry to adopt DECOAT solutions
Video production, use of social media, create public awareness</td>
<td rowspan="2">Sales channels
Already established partners’ network in their countries and across EU
Industrial exhibitions
Direct use from Industrial partners
Licensing agreements</td>
</tr>
<tr>
<td>Business model
Materials: Direct use by the Industrial partners;
Processes: licencing agreements with related industries
Equipment: direct sales of optical monitoring and recycling equipment</td>
</tr>
<tr>
<td>Budget and funding:
Financial summary:
The initial target will be to license the DECOAT recycling process know-how to recyclers and to sell the DECOAT recycling technology.
Funding required:
Further development costs need to be covered (eg via H2020).
Further investments:
Needed to increase the TRL of DECOAT technology to market readiness, via combination of public and private funding</td>
<td>Novel market creation for DECOAT outcome
 
</td>
<td>Revenue channels (after 7yrs)
Recycling activities of coated parts (€140mio/yr)
Triggerable additives sales: microcaps, carbon-based nanomaterials,.. (€20mio /yr)
Recycling equipment sales: advanced optical monitoring and sorting; MW oven, Creasolv systems (€35mio total)
Recycling Equipment servicing: maintenance, technical consultancy (€7mio/yr)</td>
</tr>
</table>

E. Training development

Audio-visual material will be prepared from the development work and demonstration activities. The material will be used to train personnel from the company partners and public material will be put online (interactive way) to improve skills and competences of persons working in (mainly) the plastics, textiles and recycling industry. Workshops hosted at the pilot plant set ups at AIMPLAS and IVV, among others, will be organized to promote DECOAT foreground. The workshops will act as multipliers of dissemination efforts.

F. Standardisation and Regulatory requirements

Participating in standardisation activities will enable DECOAT to: (i) gain intelligence on standards in areas touched by DECOAT, e.g. recycling, sustainability, carbon footprinting, end-of-life and safety; (ii) contribute to the development of new standards and ensure that specific needs are taken into account; (iii) establish contacts with other stakeholders, experts and regulators at national, European (CEN) and worldwide (ISO) levels. See also Task 7.4.

G. Evaluation of the PEDR

The plan will also envisage ways of **measuring the dissemination and exploitation efforts and its impact on DECOAT partners and the stakeholder community**. We will continuously pay attention to this but also do explicit evaluations before starting the DC2 and DC3 action so that the PEDR can still be adapted for the final DECOAT part, see WP7 (Task 7.1).

Management of research data

Data and Knowledge produced during the DECOAT project will be handled within the tasks of the Data Management Plan. DECOAT will also join the Open Data Pilot. A data repository will be created in order to store raw data, results, reports and deliverables produced during the project. During the early stages of the project, partner IRES will discuss with partners and take a decision about the accessibility of the data. A justified decision on which simulated and experimental data will be confidential and which will be public will be made. As Data Repository, one of the European Open Access is foreseen.

Raw data generated by instruments during experiments will be open and shared with the public. Confidential and sensitive data related with industrial processes or products will be restricted and shared only among the Consortium partners. In matters of reusability, not sensitive or raw data will be shared among the research community through existing web Data Repositories such as the UCI repository[93], which is widely used for Materials Data. Access rights and intellectual property will be managed according to the grant agreement and the consortium agreement. The kinds of data generated by the project are described in the following table.

Research Data produced	Accessibility	Data Exploitation, Accessibility and Reusability
Experimental Procedures	Public	Reports, publications
Experimental Data	Consortium	Storage and preservation in data repositories
Data for Economic and Environmental Impact	Public or Consortium	Storage and preservation in data repositories and possibly public or private reports
Sensitive Industrial Data	Consortium	Storage and preservation in data repository (private)

In terms of Data Exploitation and Reusability, partners working towards publishing their work (articles, monographs, conferences etc., preferably in high impact journals) are committed to comply with the "Open Access" guideline of the DECOAT project. More specifically, published work should be easy to access, online readable, easy to download and print. This can be done by sharing the above mentioned publications into online archives and providing open access to them such as Open Access Infrastructure for Research in Europe (OpenAIRE) and the Directory of Open Access Repositories (OpenDOAR).

Knowledge management and IP protection

Knowledge management to ensure open access. DECOAT project is focused on 'research & innovation' (this being a RIA) so a fair number of scientific articles is expected (see Table pag 32). For these publications, publishable results will follow an open-access model in order to optimize the impact of publicly funded scientific research, both at European level and at Member State level. For what concerns other dissemination activities (Website, Workshops, Conferences and Fairs) **DECOAT is fully committed to ensure full dissemination and high transparency of the data and results to ensure "public access"** to project results. **Only limitation is that it should be in line with the IPR management**. We will make for example large parts of the environmental and economic evaluation (from WP6) publicly available. The strategy is clearly to provide as much as possible the knowledge generated by DECOAT to relevant communities for free, e.g. via partner ISWA (NL) and via ASG partners (like EUPC, EuMaT, European Coatings) as this will be beneficial for further collaboration and for levering the DECOAT impact.

93 See: https://archive.ics.uci.edu/ml/datasets.html

IPR Management during DECOAT. Management of knowledge, intellectual property and of other innovation related activities involves the recognition of project results/innovations that may have commercial value, the protection of novel materials and processes developed in DECOAT and the determination of ownership of technologies and resolution of ownership conflicts. For the success of DECOAT all project partners agree on explicit rules concerning IP ownership, access rights to any Background and Foreground IP for the execution of the project and the protection of intellectual property rights (IPRs) and confidential information before the project starts. Such issues will be addressed in the Consortium Agreement that will be developed in line with the Grant Agreement (cf. H2020 General model grant agreement).
Consortium Agreement. The purpose of the Consortium Agreement is to establish a legal framework for DECOAT in order to minimize any internal issues during the DECOAT implementation. It will be based on the DESCA 2020 Model Consortium Agreement regulating albeit general definitions and purpose through descriptions responsibilities of participating parties, project entry, duration and project termination, as well as mutual liabilities.
Ensuring freedom to operate. For the preparation of the DECOAT proposal a patent analysis was made, see section 1.4.2. This showed that for implementing DECOAT there is freedom-to-operate. During DECOAT implementation, the external IP situation will be monitored as part of the exploitation activities (see Task 7.2). If forced to by the appearance of an external patent, mitigation actions will take place leading potentially to an adjustment of the scope of DECOAT.
Access Rights to Background and Results during the project. To ensure a smooth execution of the project, the DECOAT partners agree to grant royalty-free Access to Background and Results IP for the project implementation. All project partners will determine any Background IP they are willing to submit to DECOAT within the Consortium Agreement before the project starts.
IP Ownership within the consortium. IP on Results shall be owned by the project partner carrying out the work leading to such IP. If any IP is created jointly by at least two partners and it is not possible to distinguish between the contributions of each of the project partners, such work will be jointly owned by them. Any such joint inventions and all related patent applications and patents shall be jointly owned by the contributing parties.
External Non-Disclosure Agreement (NDA). Any external party interested in testing materials developed within DECOAT during the project must write a LOI (Letter Of Intent) to the consortium, indicating the intent and purpose of their contribution. Each LOI has to be approved by the consortium and the interested party has to sign an NDA.

(b) Communication activities

The DECOAT consortium will promote the project and its results, by providing targeted information to multiple audiences, in a strategic and effective manner and engaging in one- and two-way exchange. The communication activities will start at the onset of the project and continue throughout its entire duration. The communication strategy has the **key objective to show the benefits of DECOAT results to the European coating, plastics and recycling communities**.
The communication strategy has been delineated to ensure a high visibility of the project and to help maximize the impact of results. We follow the guidelines from the Lead Market Initiative Ad-hoc Advisory Group for Bio-based Products[94]: "*Design and implement a communication strategy involving all partners in the value chain and all other stakeholders to achieve coherent messages on bio-based products. (COM/9)*". Detailed description:

- *Goal*: firstly to reach out to the general public to create positive momentum for recycling of materials (especially as their collaboration will be needed to collect certain items like EoL textiles) and, secondly, make them aware of the impact of EU collaborative projects;
- *Method*: using story-telling to make the matter easier accessible and to show the human face of the involved people
- *Audience*: wider public as coated products are being bought all the time by all the people. As such our back pack will be used as an example close to public.
- *Key messages*: importance to select recyclable by design products as this will have a life cycle impact.
- *Channels*: use of press release in combination with popular media is seen as the most effective to reach a larger public. This will be backed up by use of social media (YouTube, Twitter,…) and the project website and available public material, accessible for an (interested) layman. These outreach activities will be carried out at EU, national, regional and local level, under the responsibility of the local partners, but coordinated by the dissemination manager.

Communication activities will support the key dissemination moments that are formed by the three Dissemination Campaigns (see part C of PEDR) as that should be excellent moments to reach a wider audience and to focus on user acceptance in turn generating further demand for recyclable products.
We will also interact with other European (H2020) projects and relevant European networks, see Task 7.5.

94 See: LMI Ad-hoc Advisory Group for Bio-based Products – List of Priority Recommendations 2011, see: https://biobs.jrc.ec.europa.eu/sites/default/files/generated/files/policy/2011%20Lead%20Market%20Initiative%20LMI%20Biobased%20Products%20Priority%20Recommendations.pdf